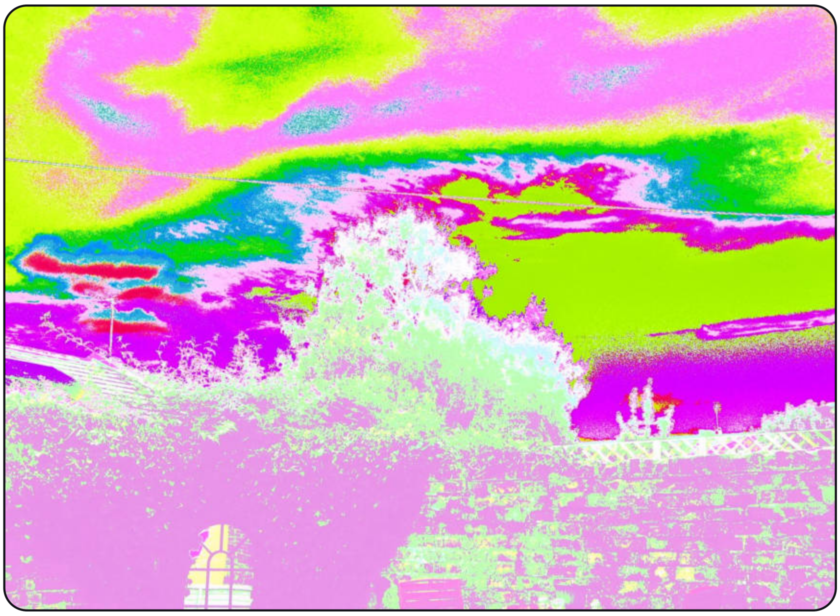
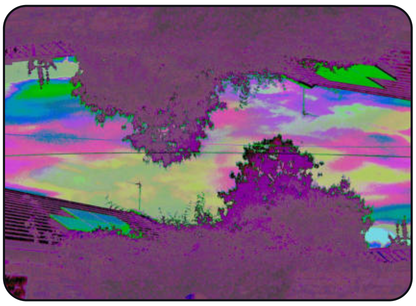
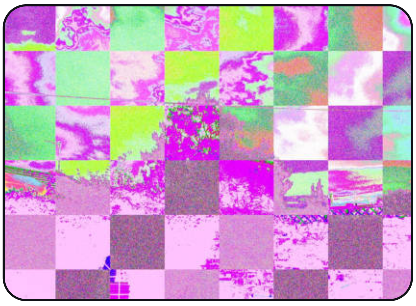
CHDK REWIRED

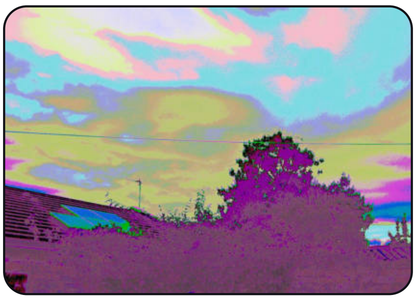
Instruction Manual/Reference

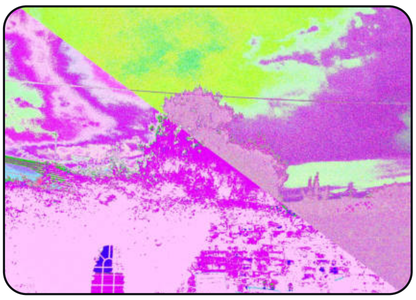
A field guide to shooting a software-circuit-bent Canon PowerShot. Written around the Canon Powershot A470; but every other supported body works a similar way, and where one does not, this is highlighted.











REWIRED
OPTICS

@rewired_optics

CONTENTS

Chapters one to four are the two bend engines and the way they are cut up; five is multiple exposure, six the Game Boy, seven how to use any of it on purpose. The front three sections are the ones to read before touching the camera.

THE THIRTY-SECOND VERSION

This camera is just an ordinary Canon PowerShot with a piece of custom software loaded off its memory card. The software sits between the ADC and the JPEG and rewires what passes through: it can reroute the data lines coming off the ADC, imitate memory and timing faults, cut the frame into regions that are each wired differently, stack several exposures into one file, and run Game Boy games. Take the prepared card out and the camera is stock again.

ADC ▶ MATRIX ▶ SEGMENTS ▶ PROFILES ▶ EXPOSURE ▶ JPEG + DNG

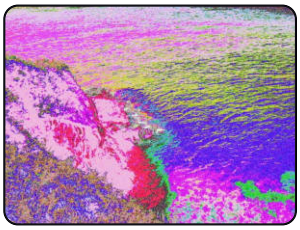
Everything in this manual happens in the middle of that chain, inside the camera, before it even writes the file. Each chapter opens with this bar and its own stage lit.

BEND MATRIX	Rewire the ADC data lines: swap them, invert them, duplicate one across several, or tie one high or low.
SEGMENTS	Give different parts of one frame different matrices - sixteen layouts, from halves to Truchet tiles.
PROFILES	Named failures of the memory downstream: stuck address lines, mis-set DMA stride, missed refresh, a foreign buffer on the data lanes. Up to four chained.
MULTIPLE EXPOSURE	Combine two to nine frames with Additive, Lighten, Darken or Average, with a ghost of what you have so far held over the live view.
RECIPE RECALL	Save numbered bends to the card - or load the exact recipe back out of a photograph you liked.
GAME BOY	Play `.gb` / `.gbc` ROMs. The Game Boy Camera cartridge looks through the real Canon lens.
DRESSING	Five themes, a set of custom grids and reticles, and shutter, button and self-timer sounds you supply as `.wav` files.

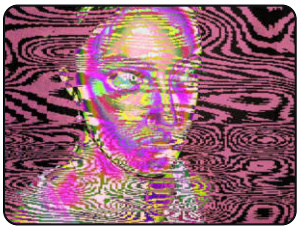
TRY THIS

Hold PRINT for a second to open the patchbay, press MENU, and load one of the saved bends. Take a photograph. You cannot damage the camera by making an ugly bend - the worst case is a slow save and a picture you delete.

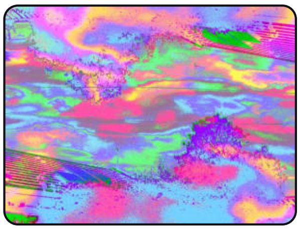
WHAT IT LOOKS LIKE ON THE WAY IN



matrix bend



Warped lines size 55%



matrix bend

Centre circle size 50%

HOW TO READ A STATE STRIP

Most photographs in this manual carry a strip like this one underneath them. It is not decoration: it is that frame's own recipe, decoded from the JPEG comment the camera wrote into it. Ten cells, pin 9 at the left, each showing where that outgoing line got its value.

d9	d5	~8	L0	B3
straight - unbent	rerouted	inverted	tied to a rail	a non-data bus

A CHAIN line under the strip names the experimental profiles that ran after the matrix; a SEGMENTS line names the layout that cut the frame up. A photograph with no strip at all was taken straight.

Two of these cells read differently on the camera. A bus cell is `B3` here and the bus's own two-letter name - `OB`, `NZ`, `HC` - on the camera, which has ten columns across 360 pixels and no room for more. And the strip is twelve cells wide, not ten, on any frame shot with the A480.

The shooting screen

The useful readings stay around the live view:

- **Top left:** the live histogram.
- **Top centre:** `EXP 1/2` - the exposure about to be taken, in a multiple-exposure sequence. Nothing is drawn here when no sequence is running.
- **Top right:** the bend plate. It alternates every 2.5 seconds between the matrix - ten source and destination pins with their routes between them, `SAVED BEND02` or `UNSAVED bend` named under it - and two lines in its place: `X` for the experimental profile or `chain of 2`, and `SEG` for the segment layout. Both lines are always present and say `off` when the engine is not running.
- **Bottom left:** battery charge and voltage, ISO, shutter speed, free card space and free working memory.
- **Bottom right:** JPEG or DNG mode, shots remaining, body/sensor temperatures, flash and focus mode, zoom step, equivalent focal length and focus distance.

WHAT CHDK IS

CHDK is the **Canon Hack Development Kit**: a long-running, open-source project that adds features to supported Canon compacts without replacing Canon's firmware. This edition keeps that foundation and adds two bending engines, segmented bends, in-camera multiple exposure, a Game Boy module, and a rebuilt on-screen interface with themes and custom sounds.

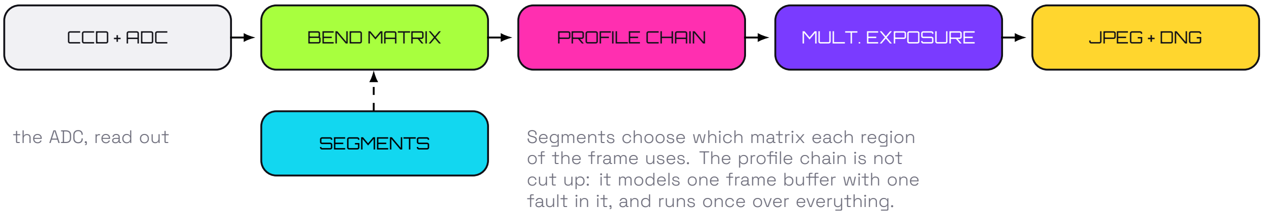
At power-on the camera reads `DISKBOOT.BIN` from the prepared card into memory and runs it. Canon's own firmware is never modified: nothing is written to the camera, and the whole build lives on a card you can pull out.

ON THE CAMERA

Turn the camera on normally with the supplied, locked card in it. Wait for the custom boot screen - or, on the bodies that have none, for the overlay to appear over the live view - then use it as a camera. To get a stock camera back: power off and swap in an ordinary, non-bootable card.

WHERE THE BEND ACTUALLY HAPPENS

Both engines run inside the raw hook - after the sensor has been read out and before Canon develops the JPEG. That timing is what makes these bends behave like circuit bending rather than like a filter: the result is baked into the JPEG and into the DNG if you shoot raw, because nothing was ever applied to a finished picture.



GOOD TO KNOW

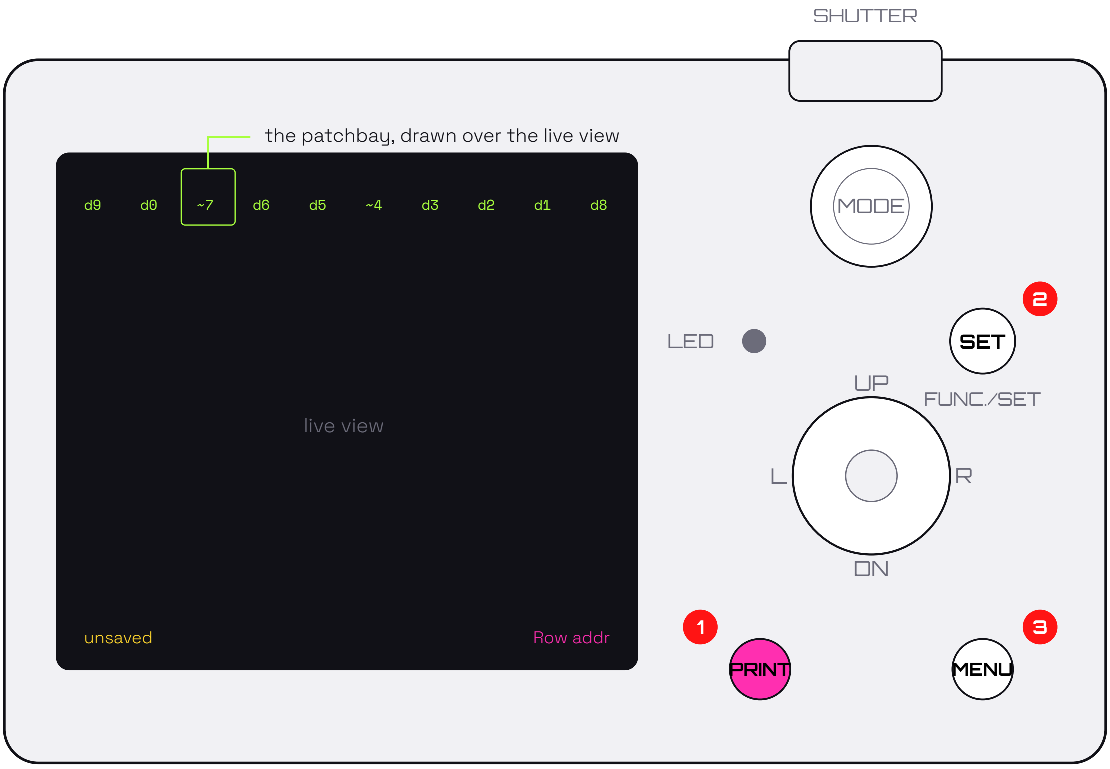
CHDK is unofficial software and not a Canon product. Do not remove the battery or the card while the blue activity light is on - that is the camera writing, and a bent frame can take a long time to write, especially with several effects chained.

WHAT IS ON THE CARD

DISKBOOT.BIN	The build itself, loaded into memory at power-on. The card's write-protect tab is what tells the camera to look for it (the little SD card switch).
CHDK/BENDS/	Your saved bends, `BEND00.BND` upward. Back this folder up before you ever format the card.
CHDK/GBC/	Game Boy ROMs you supply, with `SAVE/` for cartridge saves and `PHOTOS/` for the Game Boy Camera album.
DCIM/100CANON/	Your photographs, exactly where Canon puts them, with the bend recipe inside each JPEG.
DCIM/GBC/	Game Boy Camera pictures exported as numbered bitmaps, so they come off the card with the rest of the shoot.
CHDK/SOUNDS/	`shutter.wav`, `button.wav`, `selftimer.wav`. Supply any or none - a missing file leaves Canon's own clip in place.
CHDK/GRIDS/	Framing grids and gun-sight reticles, chosen from the menu.

THE CONTROLS YOU NEED

The A470 has ten keys and no zoom rocker, and that shapes almost everything below. UP and DOWN are the zoom on the shooting screen, so the patchbay cannot have them and is opened by **holding the ALT button** instead. The camera focuses and shoots normally throughout, patchbay open or not.



1 PRINT IS ALT

Tap it for the CHDK menu short-cut. **Hold it one second** to open the patchbay, and hold it again to leave.

2 FUNC./SET

In the patchbay, **L/R** move the cursor along the pins and **SET** - this button, not the middle of the pad - arms the one under it. **UP/DOWN** then choose its source.

3 MENU

Opens the browser inside the patchbay. Everywhere else it is the CHDK menu, and the way back out of anything.

HOLD PRINT	Open or close the live bend patchbay. The same hold both ways - there is no separate exit.
MENU	Inside the patchbay: open and close its `SAVED` / `EXPERIMENTAL` / `SEG` browser. Elsewhere: the CHDK menu, and Back.
LEFT / RIGHT	Move the highlight along the pin strip. In the browser, change window; on an indented Size or experimental-control row, adjust that value instead.
SET	Choose and confirm. On the strip it first arms a pin; SET again keeps the change, MENU cancels it.
UP / DOWN	Walk a source list or a menu. On the shooting screen, outside the patchbay, they are the zoom.
SHUTTER	Half-press to focus, full to shoot - and the shutter stays live while the patchbay is open. You do not dismiss anything to take a picture.
PLAYBACK	Review images. The CHDK menu can load a bend recipe out of the image you are looking at.

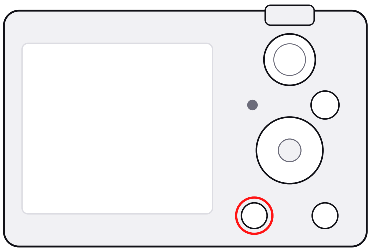
IF YOU ARE HOLDING AN A480

Everything in this manual is true of the A480 except the four rows below. It has a zoom rocker and a playback key that the A470 does not, and the build uses both.

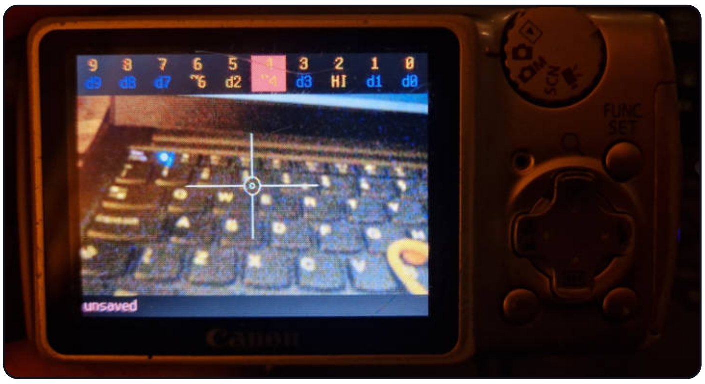
OPENING THE PATCHBAY	UP on the shooting screen opens it, and UP closes it again. The one-second ALT hold still works and is still the only route on the A470. Canon binds nothing to UP on this body, which is why it was free.
THE ZOOM ROCKER	On the strip, it walks the saved bends: ZOOM IN loads the next one, ZOOM OUT the previous. Editing or randomising detaches the live matrix; the next rocker press picks the sequence up again.
THE TUNING KNOB	In the browser, highlight an indented Size or experimental-control row and the rocker turns it. On the A470 that job moves onto LEFT/RIGHT on the same row.
TWELVE PINS	Its ADC reads twelve bits per pixel, so the strip is twelve cells wide instead of ten. Nothing else changes, and a bend saved on one body loads on the other.

The A410 and A460 sit with the A470: ten keys, ten pins, no rocker, ALT hold to open the patchbay. All three have a DISP key the A470 lacks, which matters only for the Game Boy - see chapter six.

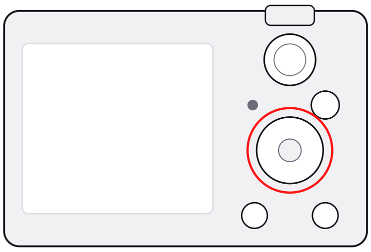
The patchbay, open



Held PRINT for a second on the shooting screen. The strip across the top is the ten data lines - pin number above, where that pin is fed from below - and the live view carries on underneath it. `unsaved` at the foot is the status line.

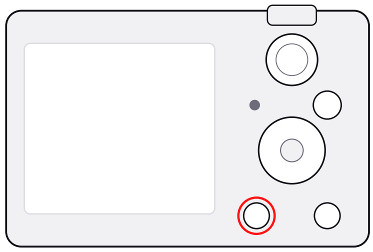


A pin under the cursor



LEFT and RIGHT walk the highlight along the strip; here it is on pin 4. Nothing has changed yet - the cursor only chooses which pin SET will arm.

The bend plate, patchbay closed



The bend state stays on the shooting screen after you leave the patchbay. The plate alternates every 2.5 seconds between the matrix graphic and these two lines: `X off` means no experimental profile is running, `SEG off` that the frame is not cut into regions. Both lines are always drawn, so the answer is always in the same place. `1/2` at the top is a multiple-exposure sequence waiting for its first frame.

If you get lost: press MENU until panels close, then hold PRINT to leave the patchbay. A half-finished multiple exposure is the one thing that does not clear this way - use **Multiple exposure ▸ Cancel sequence**.

TRY THIS

Take one straight photograph before you start. It is your control frame, and it proves the light, the focus and the card are all good before anything is deliberately broken.

CHAPTER ONE

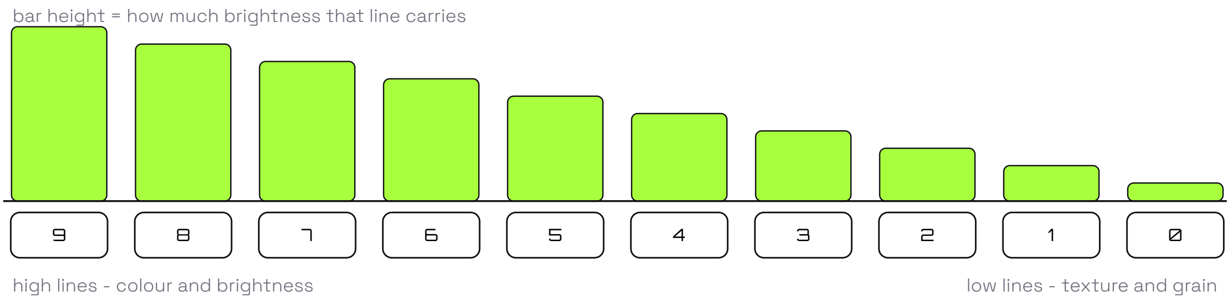
THE BEND MATRIX

Ten invisible patch cables between the ADC and the processor. You are holding both ends.

ADC ▶ **MATRIX** ▶ SEGMENTS ▶ PROFILES ▶ EXPOSURE ▶ JPEG + DNG

HOW A DIGITAL BEND WORKS

Every pixel leaves the ADC as a ten-digit binary number. The high digits carry large steps of brightness; the low digits carry fine detail. The matrix decides, for each outgoing digit, where its value comes from - and it does this before Canon turns that data into a picture, which is why the result is in the file rather than on top of it.



Rewire a high line and the picture changes character completely: a face can go metallic, a sky can invert. Rewire a low line and the subject stays recognisable while its texture roughens. Most bends worth keeping do a bit of both.

WHAT A PIN CAN BE PATCHED TO

`D0–D9`	Another data line, used as it is. `d9` on pin 9 is the straight, unbent connection.
`~0–~9`	The same line, inverted - read on the wrong polarity, as if pulled through an inverter.
`LO / HI`	The line shorted to ground or to the rail. It stops carrying anything and holds still.
BUSES	Twelve of them: the pixel and line clocks and their divisions, frame and diagonal ramps, the optical black reference, noise, sample-and-hold, line memory, bus parity and a tap. Locked away behind **Simple signals only** because they are slow - see the note overleaf.

THE STRIP, AS THE CAMERA DRAWS IT

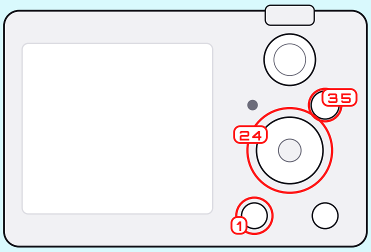
The patchbay is drawn over the live view, in the overlay plane, so it stays rock steady while the picture under it moves. Pin 9 is at the left. The cursor is the box; SET arms the pin under it and only then do the arrows change anything.

The line along the foot has three fields, and each is blank when it has nothing to say. **Left:** `BEND07` when the live matrix is still the file's, `unsaved` once you have touched it, `straight` when nothing is patched. **Centre:** the segment layout and which region the pins are editing. **Right:** the experimental profile by name, or `chain 3`, with a * when what is running is one of the slow ones.

the cursor - pin 7, patched to line 7 inverted

ON THE CAMERA

Hold PRINT to open the patchbay. LEFT/RIGHT choose a pin. SET arms it, UP/DOWN walk the source list, and SET again keeps it - MENU cancels that one edit. Shoot whenever you like; the shutter never stopped working.



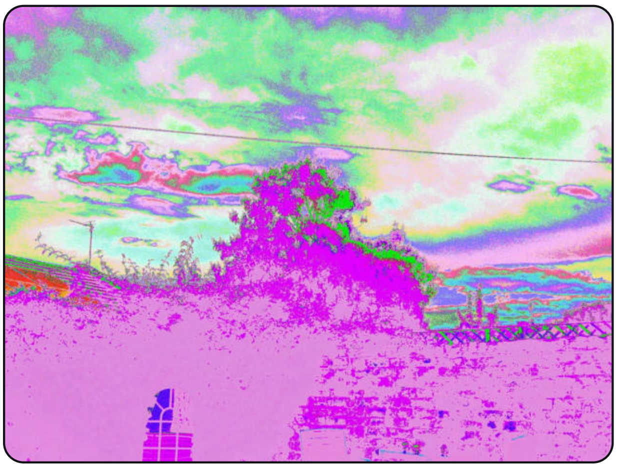
RECIPES WORTH STARTING FROM

SUBTLE TEXTURE	Swap two low pins, such as 0 and 1. The picture stays itself and its grain changes.
SOLAR COLOUR	Invert one middle or high pin - 7 is a good first choice - and leave pin 9 straight.
HARD POSTERISATION	Tie a middle pin to `LO` or `HI`. Levels collapse into bands.
RECOGNISABLE CHAOS	**Randomise** at depth 2 or 3, with **Simple signals only** left on.
FULL SCRAMBLE	Reverse or rotate the whole matrix, then **Mutate one edge** repeatedly until an image comes back.

Random depth is how many patch leads the randomiser moves - not how strong an effect is. **Mutate one edge** changes exactly one connection, which is the tool for a result that is nearly right. **Reset to straight** puts every line back on itself.

Recipe none recorded

The control frame. Every line on itself, nothing patched - this is what the scene actually looks like, and it has no recipe because there was no bend to record.



Matrix 10 data lines • random depth 3 • trash White

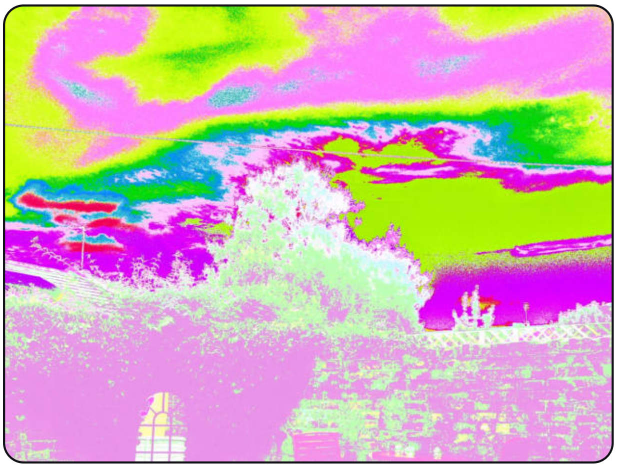
9	8	7	6	5	4	3	2	1	0
d9	d5	d8	d6	d7	d4	d3	~2	d1	d0

One inversion and a three-way ring. Pin 2 is read on the wrong polarity; pin 8 takes line 5, pin 5 takes line 7, pin 7 takes line 8. Everything else is straight.

Matrix 10 data lines • random depth 3 • trash White

9	8	7	6	5	4	3	2	1	0
d9	d0	~7	d6	d5	~4	d3	d2	d1	d8

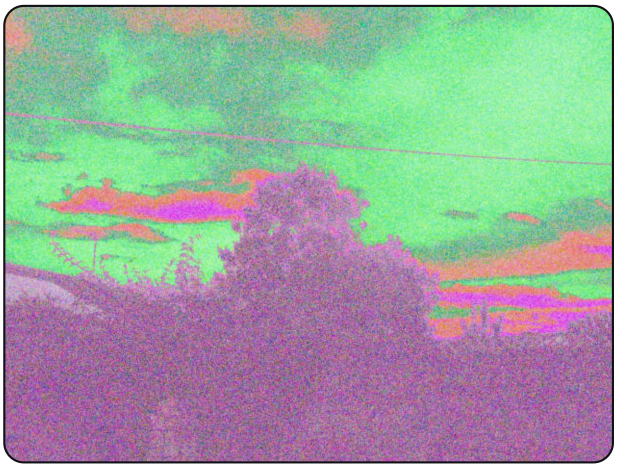
Two inversions and a swap across the whole width of the bus - pin 8 fed from line 0, pin 0 from line 8, and 7 and 4 inverted.



Matrix 10 data lines • random depth 3 • trash White

9	8	7	6	5	4	3	2	1	0
d6	~8	L0	d9	d5	d4	d3	d2	d1	d0

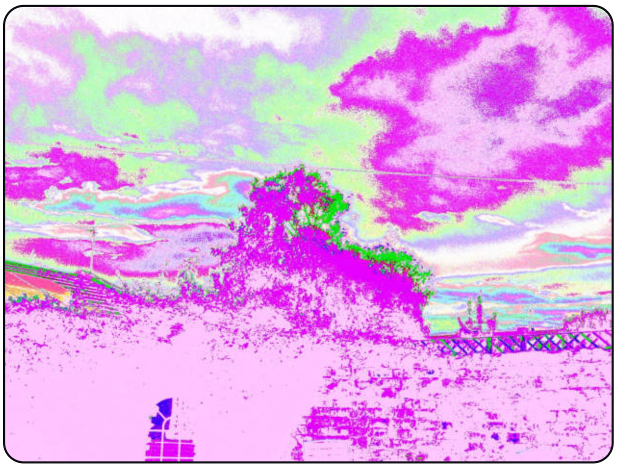
Pin 9 fed from line 6, pin 6 fed from line 9, pin 8 inverted and pin 7 tied L0. Touching the top of the bus rewrites the palette outright.



Matrix
10 data lines • random depth 3 • trash White

9	8	7	6	5	4	3	2	1	0
d9	d1	d7	d6	d5	d3	d4	d2	d8	HI

Pin 0 tied `HI`, pins 4 and 3 traded, and pins 8 and 1 traded across the bus. Tying the lowest line is the gentlest thing on this page - it removes one step of fine detail and little else.



Matrix
10 data lines • random depth 3 • trash White

9	8	7	6	5	4	3	2	1	0
d5	d8	d7	d9	d6	d4	d0	d2	d1	d3

A rotate: five lines move at once and nothing is inverted or tied. The scene is intact and the palette is somebody else's.

TEN LINES, ON A REAL SUBJECT

Everything so far was shot at a fixed scene, on purpose - an A/B test needs a scene that holds still. This is the same engine, the same ten lines, pointed at something worth photographing. These frames are from the **A410**, the smallest body the build runs on, at random depth 6 - the randomiser moving six of the ten patch leads on every press. Count the strips and some carry more than six non-straight cells: moving one lead onto another displaces that one too.

Matrix
10 data lines • random depth 6 • trash Ramp

9	8	7	6	5	4	3	2	1	0
d9	d7	d0	d1	d3	d4	d5	d2	d6	d8

Body
Canon PowerShot A410

Seven lines rerouted and nothing inverted or tied. The headland, the water and the sky are all still exactly where they were - only the meaning of each brightness step has changed.

Matrix
10 data lines • random depth 6 • trash Ramp

9	8	7	6	5	4	3	2	1	0
d9	d8	d5	~6	~7	d2	d1	d4	d0	d3

Body
Canon PowerShot A410

Two inversions in the middle of the bus. Inverting a line does not shift the picture, it folds the range, which is why the sea has gone to a colour that was never in it.

Matrix 10 data lines • random depth 6 • trash Ramp

9	8	7	6	5	4	3	2	1	Ø
d9	~6	d0	d8	d1	d5	d3	d4	d2	d7

Body Canon PowerShot A410

Eight leads moved and pin 8 inverted, with pin 9 left straight - so the broad light and dark of the scene survives underneath the colour.

Matrix 10 data lines • random depth 6 • trash Ramp

9	8	7	6	5	4	3	2	1	Ø
d9	d6	d7	d5	d0	d4	d3	d2	LO	d8

Body Canon PowerShot A410

One line tied `LO` and four rerouted. Tying a line removes a whole step of brightness, which is what flattens the water into bands.

TRY THIS

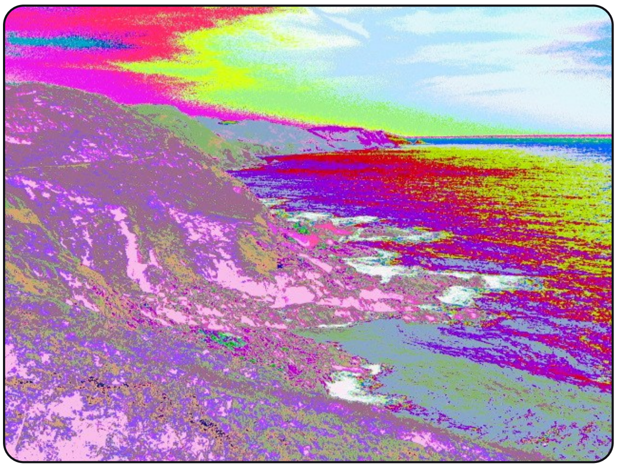
Hard horizons, water and sky are the best subjects on this page and it is not an accident: a bend remaps brightness, so a scene built from large smooth gradients turns those gradients into bands you can actually read. A cluttered subject hides the same bend completely.

Matrix 10 data lines • random depth 6 • trash Ramp

9	8	7	6	5	4	3	2	1	Ø
d7	d8	d3	d6	HI	d4	d9	d2	d1	LO

Body Canon PowerShot A410

Pin 3 fed from line 9 - the top of the bus driving one of the bottom lines - with pin 9 taking line 7, pin 5 tied `HI` and pin 0 tied `LO` underneath it.



Matrix 10 data lines • random depth 6 • trash Ramp

9	8	7	6	5	4	3	2	1	Ø
d9	~6	~7	d8	LO	HI	d0	d2	d1	d3

Body Canon PowerShot A410

Two inversions and both rails in use, one line tied `LO` and one `HI`. This is close to the limit of what still reads as a photograph.

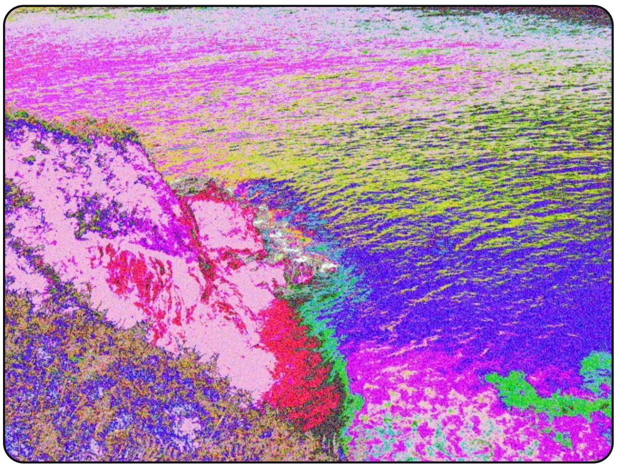
AND PAST THE LIMIT

Matrix 10 data lines • random depth 6 • trash Ramp

9	8	7	6	5	4	3	2	1	0
d9	LO	d7	LO	~5	d4	d1	d2	~3	d0

Body Canon PowerShot A410

One line rerouted, two tied to `LO` and two inverted. The horizon is the only thing holding it together - and it is enough.

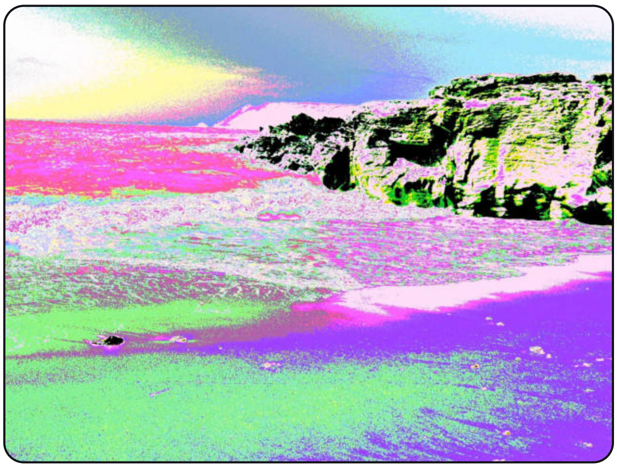


Matrix 10 data lines • random depth 6 • trash Ramp

9	8	7	6	5	4	3	2	1	0
d9	~6	d0	d8	d1	d5	d3	d4	d2	d7

Body Canon PowerShot A410

The same ten patches as the first frame on the previous page, pointed somewhere else. A recipe is not tied to a subject, which is what makes saving one worth doing.



Matrix 10 data lines • random depth 6 • trash Ramp

9	8	7	6	5	4	3	2	1	0
d6	d8	d7	d9	~5	HI	d3	d2	~1	d0

Body Canon PowerShot A410

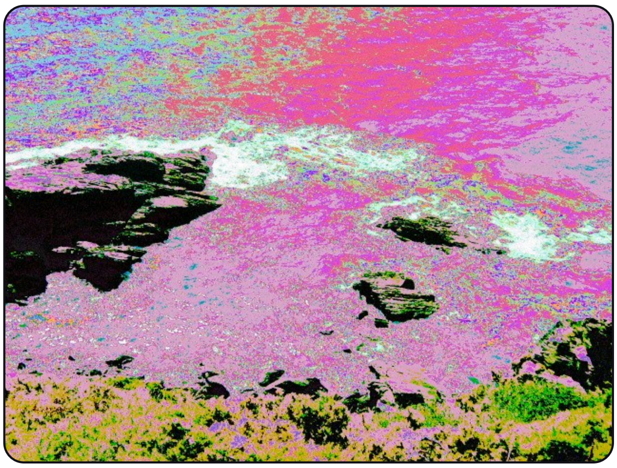
Pin 9 fed from line 6 and pin 4 tied `HI`. When the highest line is rerouted, the picture stops describing the scene's brightness at all.

Matrix 10 data lines • random depth 6 • trash Ramp

9	8	7	6	5	4	3	2	1	0
d7	d8	LO	d6	d5	HI	d3	d2	d0	d1

Body Canon PowerShot A410

And a reminder of the other direction: pins 1 and 0 traded at the bottom, two lines tied and one reroute at the top, and most of the bus untouched. The photograph is still a photograph, and it is a different photograph.

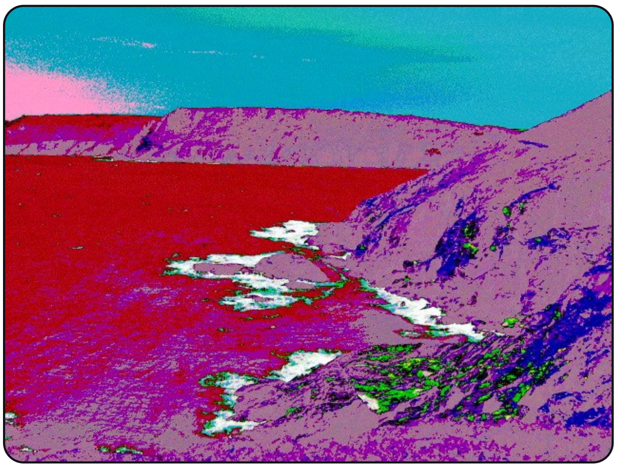


Matrix 10 data lines • random depth 6 • trash Ramp

9	8	7	6	5	4	3	2	1	0
d9	d6	d7	d8	d2	d5	L0	d0	d4	d1

Body Canon PowerShot A410

One line tied L0 and seven rerouted, with pin 9 left alone. The surf keeps its shape because the brightest lines still mean brightness.



Matrix 10 data lines • random depth 6 • trash Ramp

9	8	7	6	5	4	3	2	1	0
d9	d8	~7	L0	d2	d4	d3	~6	~1	d0

Body Canon PowerShot A410

Three inversions, two of them adjacent. Inverting neighbouring lines produces the fine banding in the water - each step of brightness crosses a fold.

GOOD TO KNOW

Simple signals only is on by default. Data lines, inverted lines and the two rails compile into a lookup table and cost nothing. Turning it off unlocks the twelve buses - and takes the capture path off its fast route onto one that touches every pixel individually. That is roughly seven and a half million round trips on this sensor, ten and a third on the A480. The picture still saves; you will wait for it. Nothing on the strip announces the slow route, so if a shot is suddenly taking seconds, this switch is the first place to look.

CHAPTER TWO

SAVE IT. FIND IT.

The camera can remember a bend on purpose - or give you back the accident you didn't know you wanted.

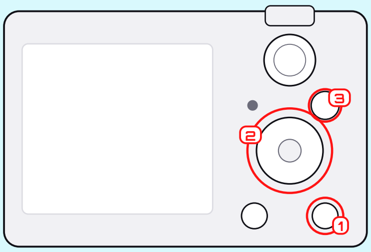
ADC ▶ **MATRIX** ▶ SEGMENTS ▶ PROFILES ▶ EXPOSURE ▶ JPEG + DNG

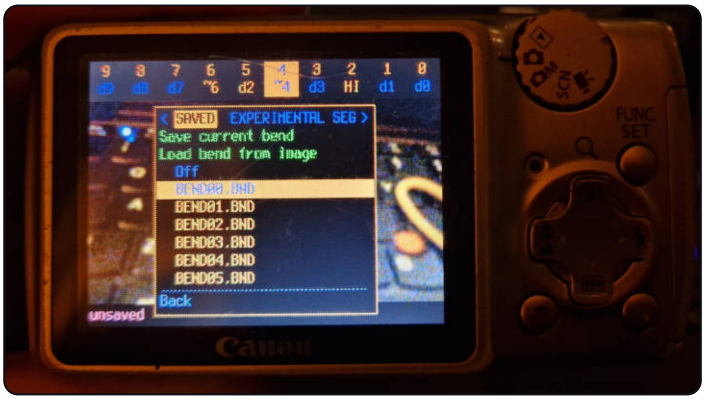
NUMBERED PRESETS

There is no on-camera keyboard worth using, so saved bends take numbered slots: `BEND00.BND` to `BEND99.BND` in `CHDK/BENDS`, of which the browser tracks sixty-four at a time. A preset holds the matrix, the profile chain, the segment layout and the random seed - everything needed to reproduce the frame.

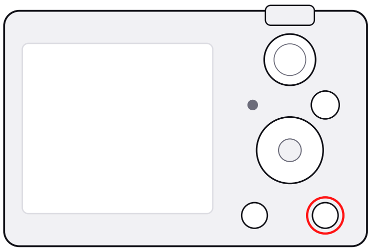
ON THE CAMERA

In the patchbay press MENU, move to `SAVED`, choose **Save**. The camera takes the lowest free number. Highlight a saved bend and press SET to load it; hold SET for its detail and actions screen.



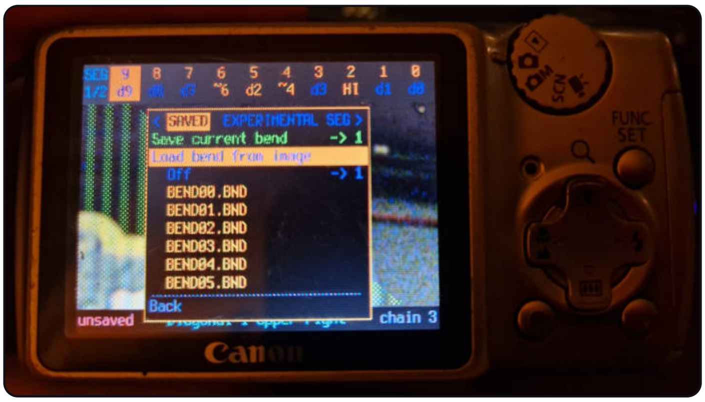


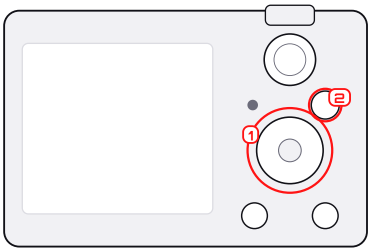
The SAVED window



MENU inside the patchbay opens the browser. Three windows across the top - `SAVED`, `EXPERIMENTAL`, `SEG` - with LEFT/RIGHT between them. Underneath: Save current bend, Load bend from image, Off, then the presets already on the card.

Saving one

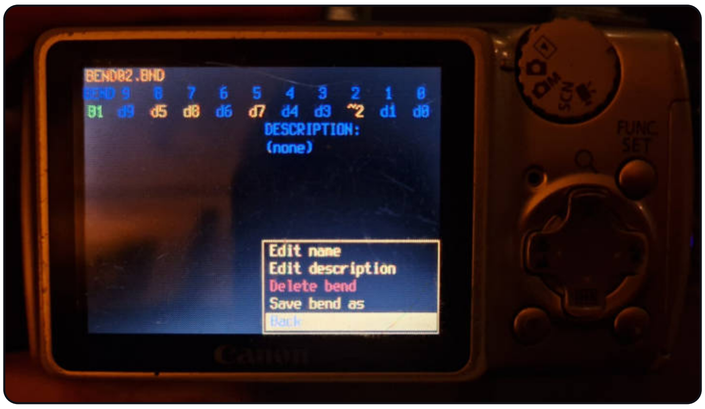


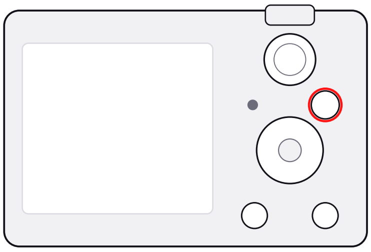


UP/DOWN to **Save current bend**, then SET. The `-> 1` at the right is the region it applies to when a segment layout is on. The camera takes the lowest free number rather than asking you to name it here.

The footer names the saved bend for as long as the live recipe still matches it. Edit anything and it changes, honestly, to `unsaved bend`. Deleting a preset removes that file only - photographs already taken are untouched.

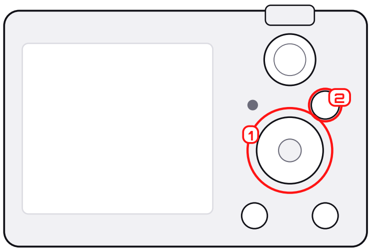
Hold SET for the detail screen





Every saved bend has one. It shows the matrix the preset holds and its description, and offers Edit name, Edit description, Delete bend and Save bend as. Deleting removes the preset file only - photographs already taken keep their own copy of the recipe.

The on-camera keyboard



Reached from **Edit name**. Arrows move around the grid, SET types the letter under the cursor, MENU cancels. It is slow, which is exactly why bends save to numbered slots first and get named later, if at all.

A named preset

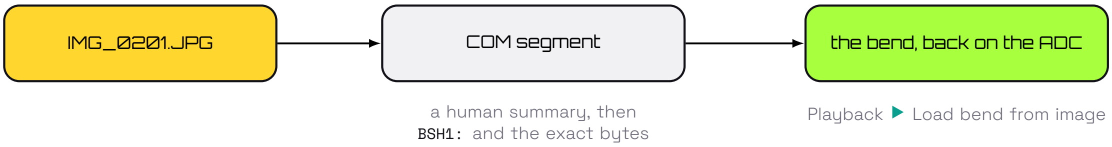
The same screen once a name and description are set: `naaeeeee [BEND02]`. The number never goes away - the name is a label on top of the slot, so a preset can always be found by the file it lives in.

THE PHOTOGRAPH IS ALSO A PRESET

TRY THIS

This is the single most useful habit in the manual. When a frame surprises you, load its recipe immediately, before the randomiser moves on, then save it to a numbered slot. Every state strip printed in this manual was read back out of the photograph above it by exactly this route.

With **Record bend with shot ▸ In JPEG** - the default - every bent frame gets a legal JPEG comment segment written in after the fact, beginning `@rewired_optics`. It holds a readable summary and then the exact machine recipe. Copy the JPEG anywhere and the recipe goes with it.



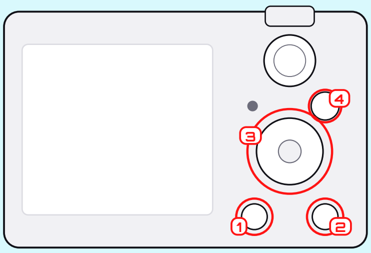
This is what that comment actually looks like, read straight out of `IMG_0201.JPG` with `strings`:

```
@rewired_optics
bend 9:d9 8:d0 7:~7 6:d6 5:d5 4:~4 3:d3 2:d2 1:d1 0:d8
depth 3  trash White
BSH1:4253483102002000034000100680000000080102031405061700…
```

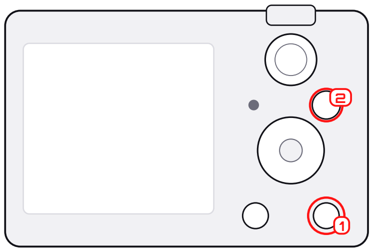
The first three lines are for you, and turn up in anything that displays a JPEG comment. The last line is for the camera: the same bytes a sidecar would hold, in hex so the whole comment stays printable.

ON THE CAMERA

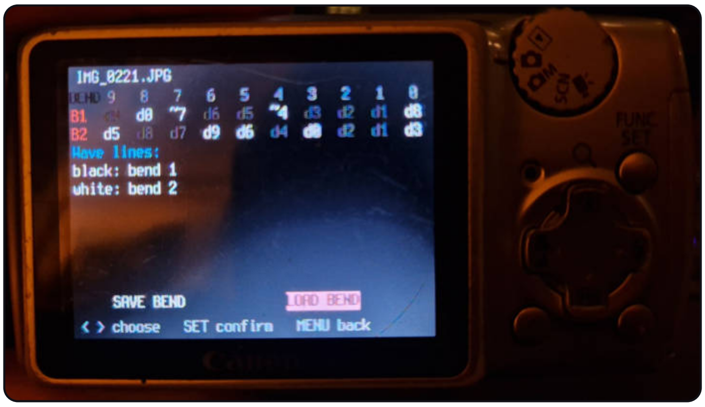
Patchbay ▸ MENU ▸ SAVED ▸ **Load bend from image**, and pick the photograph. Close the patchbay: that frame's matrix, chain and segment layout are live again. The CHDK menu ▸ Circuit Bending carries the same entry if you are already in it.



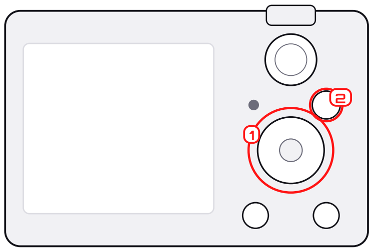
Load bend from image



The picker reads the recipe out of the photographs on the card and shows them as thumbnails, newest first. SET inspects the one under the cursor, MENU goes back. It is a separate module because the JPEG decoder it needs is far too big to keep resident.



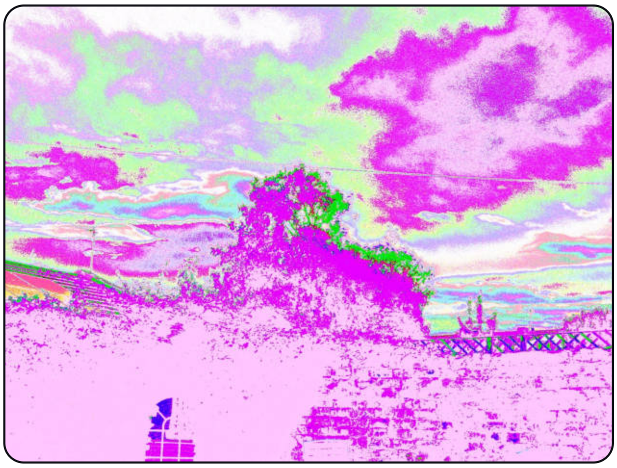
Inspecting one



The recipe that made `IMG_0221.JPG`, read back out of it: both region matrices, and the layout that decided which pixels got which - `Wave lines: black bend 1, white bend 2.` LEFT/RIGHT choose **Save bend** or **Load bend**, SET confirms.

GOOD TO KNOW

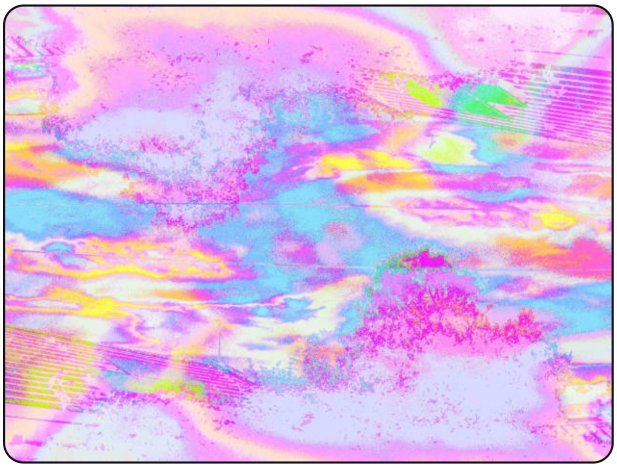
An edited or re-exported JPEG may have lost its comment - most editors drop it. **Sidecar** mode writes `IMG_1234.BND` beside the picture instead, in which case copy both files together. **Off** records nothing at all.



Matrix 10 data lines • random depth 3 • trash White

9	8	7	6	5	4	3	2	1	0
d5	d8	d7	d9	d6	d4	d0	d2	d1	d3

A rotate: pin 9 fed from line 5, 6 from 9, 5 from 6, 3 from 0 and 0 from 3. Saved to a slot the moment it came back off the card.



Matrix 10 data lines • random depth 3 • trash White

9	8	7	6	5	4	3	2	1	0
d5	d8	d7	d9	d6	d4	d0	d2	d1	d3

The last frame of a three-shot sequence taken later. Compare the strips - it is the same ten patches, loaded back and held through every exposure.

CHAPTER THREE

EXPERIMENTAL PROFILES

The matrix breaks the wires. These break the memory the wires write into - one named fault at a time.

ADC ▸ MATRIX ▸ SEGMENTS ▸ **PROFILES** ▸ EXPOSURE ▸ JPEG + DNG

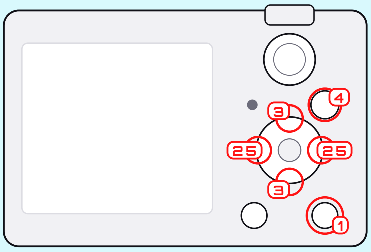
ONE FAULT AT A TIME

A profile is one named failure of the path between the ADC and the JPEG encoder: a stuck address line, a mis-programmed DMA stride, a missed memory refresh, a lost pixel clock, or something else in the camera driving the frame buffer's data lanes. They run after the matrix, because that is the physical order - the bus is rewired first, then the memory it writes into fails.

Each has an **amount** and usually a **reach**, and a star in the picker marks the ones that unpack the buffer pixel by pixel and are therefore slow.

ON THE CAMERA

Patchbay ▸ MENU ▸ EXPERIMENTAL. Pick an effect and press SET: its Amount, Reach, Data lanes, Mix and Row bands open directly beneath it. Highlight one and use LEFT/RIGHT to turn it.



The five indented rows show their current values beside their names; a dash means that particular effect does not use that control. Press SET on the effect again to fold the rows away. The full CHDK menu carries the same five controls as a fallback, but nothing is menu-only.

THE AMOUNT KNOB, AT THREE SETTINGS

Same scene, same profile, one number changed. Row addr is a stuck row address line: rows whose numbers differ only in that bit trade places, so the higher the line, the bigger the blocks that swap.

Row addr 4 amount 2

Row addr 32 amount 5

Row addr 256 amount 8

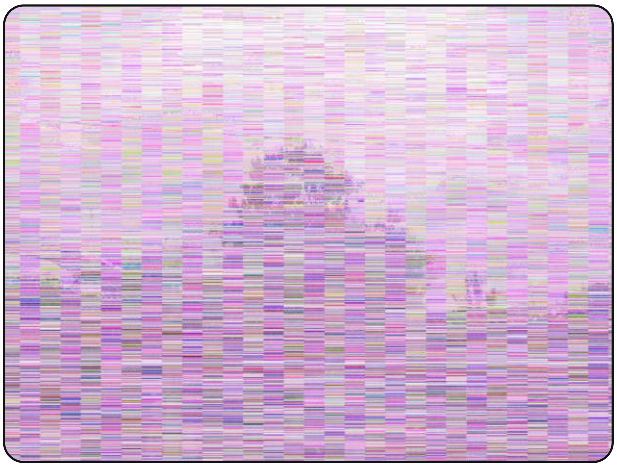
Col addr 32 amount 5

The fourth tile is the same fault turned ninety degrees: Col addr at the same setting, one column address line stuck instead of a row one. Two knobs, two axes, and nothing else changed between any of the four.

FIVE FAMILIES

MEMORY + BUS	Eleven faults of the packed bytes themselves. One linear pass, cheap enough to leave switched on.
FOREIGN BUSES	ROM or JPEG data driving the frame buffer's data lanes. They start on **XOR** rather than Replace, because at Replace they do not damage the photograph, they erase it.
PIXEL DOMAIN	Three that have to unpack every pixel, and are marked slow for it.
SENSOR PHYSICS	Blooming from charge overflowing the vertical register.
PROCESSES	Pixel sort. No camera does it; it is here because it looks good, which is a better reason than inventing hardware that would.

Overleaf is the whole list - one frame per profile, in that order - and then a glossary that takes each one in turn: what the fault is, what its two knobs really do, and what it leaves on the picture.

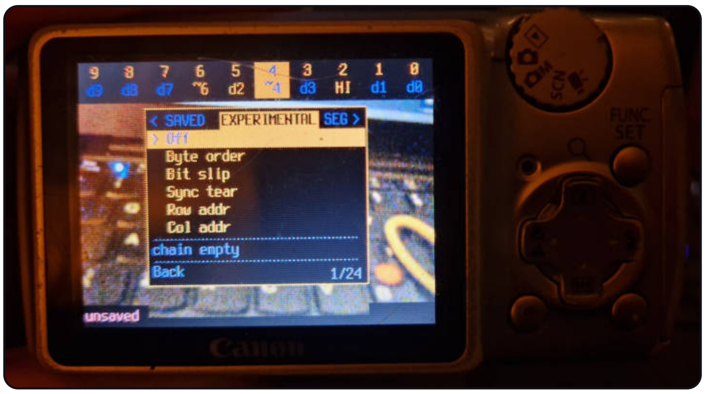


Chain **ROM bus 7** amount 7, mix XOR

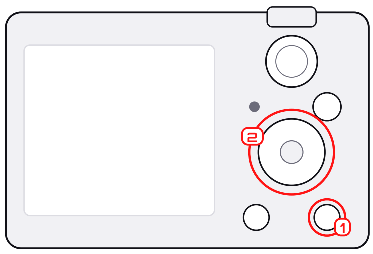
ROM bus: a megabyte of the camera's own program code - ARM instructions and Canon's string tables - driving the frame buffer's data lanes, mixed through the picture with XOR.

Chain **Refresh 8** amount 8

Refresh: missed DRAM refresh makes the packed data decay as the frame is read.



The EXPERIMENTAL window



MENU, then RIGHT to the middle window. Eighteen profiles and an `Off` entry, so `1/19` at the top. The camera lists them in the order they were built, which is the glossary's order with two exceptions: **Lock loss** and **JPEG bus** were added last and sit at the end of the list rather than with their families. `chain empty` means nothing is running yet.

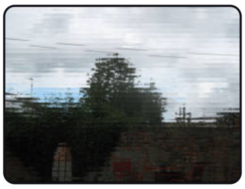
THE CATALOGUE

All eighteen profiles, one frame each, at the setting that shows it most clearly. Each label is the recipe read back out of that JPEG, not a caption anyone typed. This sheet and the glossary overleaf are both in family order, which is not quite the camera's - see the note under the EXPERIMENTAL window opposite.

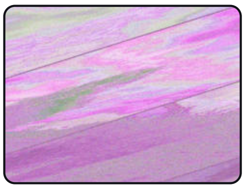

Byte order rev amount 4


Bit slip 1 amount 0

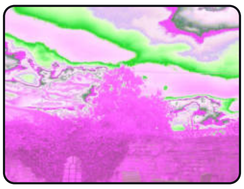

Sync tear 8 amount 7


Row addr 32 amount 5


Col addr 32 amount 5


DMA stride 4 amount 3

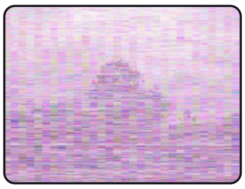

Refresh 8 amount 8

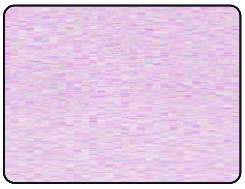

Bus echo 16 amount 15


Line mem 16 amount 14


Field flip 16 amount 14


Lock loss 8 amount 8


ROM bus 7 amount 7, mix XOR

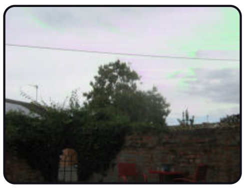

JPEG bus 7 amount 7, mix XOR


Clock slip 4 amount 4


CFA phase 7 amount 7


Clamp run 15 amount 15


Bloom 15 amount 15


Pixel sort 8 amount 8

Reading across: the memory and bus faults, then the two foreign buses, the three pixel-domain effects, Bloom and Pixel sort. Same scene, same light and the same straight matrix throughout, so the only thing that differs between any two tiles is the fault named under it.

GLOSSARY OF PROFILES

What each fault is, what its two knobs actually do, and what it leaves on the picture. Ranges are the real limits the camera will let you turn to.

■ memory + bus ■ foreign buses ■ pixel domain ■ sensor physics ■ processes

MEMORY AND BUS

Eleven faults of the packed bytes in the frame buffer. Each is one linear pass over them, so none of these costs enough to think about and any of them can be left switched on.

■ Byte order

A halfword, word or packed group latched in the wrong order - or the eight data lanes wired back to front. First tile on the contact sheet, at **rev**.

KNOBS **Amount** which reordering, of five: byte, word, group, nibble, bit-reverse. No reach.

A uniform re-texturing. The scene stays exactly where it is and only its surface changes, which makes it the gentlest thing in this list.

■ Bit slip

The row read back off its bit boundary: the bytes arrive intact and the boundaries between pixels are in the wrong place. Second tile on the contact sheet, at **1**.

KNOBS **Amount** 1–16 bits. **Reach** 0–32 whole bytes on top of that.

Because the packing is not a plain bitstream - four pixels share six bytes, out of order - this scrambles rather than shifts. Detail turns to grid; the geometry does not move.

■ Sync tear

The same slip, growing by a few bits with every row down the frame.

KNOBS **Amount** 1–16 bits per row. **Reach** 0–32, where the tear starts.

A diagonal shear that wraps: the further down the picture, the further that row has slid, until it comes back round.

■ Row addr

One row address line stuck, so rows whose numbers differ only in that bit trade places.

KNOBS **Amount** which line - 1, 2, 4 … 2048 rows apart. No reach.

Whole horizontal blocks swapped, each block internally clean. A low line interleaves neighbouring rows; a high one exchanges halves of the picture.

■ Col addr

The same fault on the other axis, at packed-group granularity so the blocks land on pixel boundaries rather than halfway through one.

KNOBS **Amount** which line - 1 to 2048. No reach.

Vertical blocks trading places. Pairs well with Row addr, which is the same fault turned ninety degrees.

■ DMA stride

The stride mis-programmed by a few bytes, so row y is fetched y times that far into the buffer instead of at the row start.

KNOBS **Amount** 1–64 bytes. **Reach** 0–15, multiplying that by 64.

A continuous shear that keeps shearing and wraps. The whole picture leans, and at large values it becomes a plait.

■ Refresh

DRAM refresh missed. Cells drift toward whichever rail their sense amp favours, and worse further down the frame - more time has passed since that part was written.

KNOBS **Amount** severity, 1–16: by the bottom row that fraction of sixteenths of all cells has decayed. **Mix** OR decays toward white, anything else toward black.

Clean at the top, increasingly eaten toward the bottom. That gradient is why it reads as time passing rather than as noise added.

Bus echo

An unterminated bus reflecting: what comes back arrives late and lands on top of what is being driven now. It reaches back past the row start into the previous row, because the scan does not stop at row boundaries.

KNOBS **Amount** 1–64 bytes of delay. **Reach** 0–15, multiplying that by 64.

A hard ghost of the picture offset to the left, laid through the picture itself.

Line mem

The line memory stops being clocked and keeps presenting the last row it latched.

KNOBS **Amount** period, 2–65 rows. **Reach** 0–15, how many rows are held each time.

Bands where the picture stops dead and one row smears down through the next several.

Field flip

Field phase inverted, so a block of rows comes back bottom-up.

KNOBS **Amount** block height, 2–65 rows. No reach.

At a block of two this is the classic odd/even line swap and reads as corrugation. At sixty it reads as the frame folded over on itself.

Lock loss

Timing losing lock and re-finding it, row by row, rather than failing steadily.

KNOBS **Amount** 1–16 slips per hundred rows. **Reach** how far it lands when it slips, 8–128 bits.

Occasional rows and runs of rows kicked sideways and then recovering. The scene stays intact with sudden slices out of place - the most broken-looking profile that leaves the picture readable.

FOREIGN BUSES

Not faults of the frame buffer but of what is connected to it: something else in the camera driving the same data lanes, which on a bench is a probe left on the wrong pad. Both start on **XOR** rather than Replace, and the reroll never puts them back on it - at Replace with every lane live they do not damage the photograph, they erase it, and nothing on the back of the camera suggests which knob brings it back.

ROM bus

A megabyte of the camera's own program ROM - ARM instructions and Canon's string tables - on the frame buffer's data lanes. Read-only on the far side, always.

KNOBS **Amount** texture scale: each ROM byte repeated 1 to 128 times. **Reach** 0–15, how fast the window walks down the frame.

Dense structured texture with visible runs and repeats in it, quite unlike noise - because it is not noise, it is code.

JPEG bus

The JPEG encoder's own 4.8 MB working buffer. It is read inside the raw hook with the pipeline stopped, so what is in there is the previous shot's compressed data.

KNOBS **Amount** texture scale, 1 to 128. **Reach** 0–15, window speed.

Entropy-coded, so it looks nothing like the ROM: high-entropy and bursty rather than structured. It is the one profile that is not live everywhere - it needs that buffer's address, which has only been recovered on the A470 and the A480. On the A410 and A460 it is a dead entry in the list rather than a crash.

PIXEL DOMAIN

These three unpack every pixel instead of walking the packed bytes, which is what makes them slow. They are also the only three that can do what they do.

Clock slip
SLOW

The pixel clock running at the wrong rate against a fixed line length, so rows stretch and compress. The rate is never wrong by a constant: it runs as a triangle, which is a phase-locked loop hunting rather than one switched.

KNOBS **Amount** rate error, 0–15. **Reach** wobble period as a shift, 0–7.

The picture breathing horizontally, wider and narrower in waves down the frame.

CFA phase
SLOW

Canon's demosaic assumes a fixed Bayer phase. Move the image under it by an odd number of pixels or rows and every red site is read as green.

KNOBS **Amount** pixels across, 0–7. **Reach** rows down, 0–7.

Geometry perfectly intact, colour destroyed. The one effect in this list that only a colour filter array can produce, and the one that looks least like a glitch and most like a broken camera.

Clamp run
SLOW

The black-level clamp loop unlocked - correcting each row against an optical black measurement taken from a row that is not there. The masked columns it reads are real masked sensor, so this is the clamp's own arithmetic done on the wrong input.

KNOBS **Amount** clamp gain, 0–15. **Reach** how far away the wrong row is, 0–15.

Horizontal banding that tracks real dark current and reset noise rather than a pattern, so it never repeats exactly.

SENSOR PHYSICS

Not a fault of the memory or the timing but of the CCD itself.

Bloom
SLOW

A well filled past capacity spills into the vertical transfer register, and everything clocked through it afterwards carries the spill.

KNOBS **Amount** how low the overflow threshold sits, 0–15. **Reach** how far the smear carries down, 0–15.

The white column standing out of every specular highlight - the most recognisable not a CMOS sensor artefact there is.

PROCESSES

Pixel sorting is not a hardware failure. It is here because it looks good, and saying so is better than inventing hardware that would do it.

Pixel sort
SLOW

Runs of pixels above a threshold, put in order along the row.

KNOBS **Amount** the threshold the runs are taken above, 0–15. **Reach** low three bits are the minimum run length; 8 or over sorts descending instead.

Horizontal streaks of sorted colour, starting wherever the picture crosses the threshold and stopping where it drops back under.

Pixel sort moves pixels in steps of two, which is not an aesthetic choice. A fault moves charge or bits and the mosaic goes along for the ride; a process moves pixels, and moving a pixel one place puts a red value on a green site. Canon's demosaic runs afterwards and assumes a fixed phase, so a one-step sort comes out as colour mud with the sort structure invisible underneath it. Two is the distance between neighbours of the same colour.

BUILDING A CHAIN

Up to four profiles, applied in order, each working on what the last one left. Order matters.

MATRIX → Row addr → Refresh → JPEG bus → Pixel sort

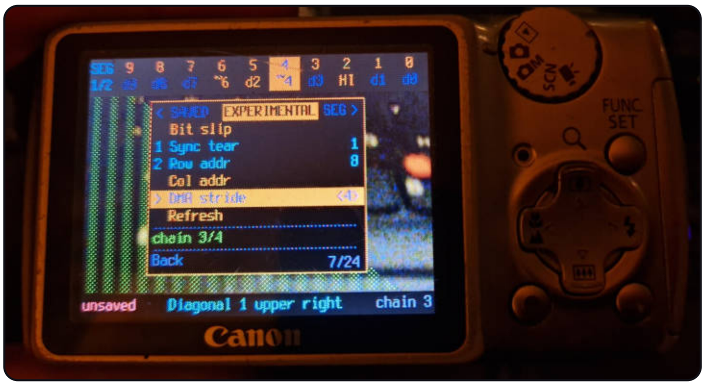
Row addr: locked · Refresh: locked · JPEG bus: locked · Pixel sort: live

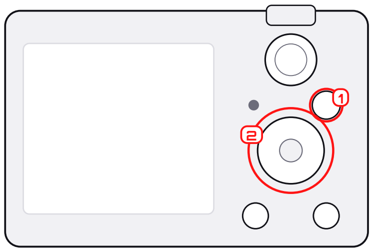
frozen - their knobs no longer move

the one the arrows are turning

Tune the live effect, then hold SET on its parent effect row: the current one freezes and a fresh live slot opens. Hold SET on a locked profile to remove it. Choosing **Off** empties the whole chain.

A chain, three deep





Two profiles are locked and numbered - 1 `Sync tear`, 2 `Row addr` - and a third is live on `DMA stride`. Pressing the live row opens its five controls immediately underneath; the selected value is in angle brackets while LEFT/RIGHT turn it. `chain 3/4` is how many of the four slots are in use.

GOOD TO KNOW

Cost adds up along the chain. Three slow profiles can mean a long wait after the shutter, during which the screen may look frozen while the blue light proves it is not. Do not power off. Data lanes, Mix and Row bands change how much of the frame a profile reaches rather than what it does; all three open beneath the effect alongside Amount and Reach.

COMBINATIONS THAT TEND TO WORK

- A geometric fault - Row addr, DMA stride - with a dirty one underneath it, such as Refresh or Bus echo.
- A colour-forming matrix first, then Bit slip at a small amount.
- A slow pixel effect last, so the fast faults before it leave it something structured.

Chain **Row addr 32** amount 5

Row addr 32: whole blocks of rows trade places, and every block is still clean.

Chain **Refresh 8** amount 8

Refresh 8: no geometry at all, just bits decaying toward a rail, worse further down the frame.

Locked in that order, the second profile receives the first one's output: the decay lands on rows that have already moved, and the block edges stay hard because Row addr ran before anything blurred them. Reverse the order and the blocks are cut out of an already-rotten frame instead - same two faults, a different photograph.

CHAPTER FOUR

SEGMENTS

One frame, one shutter press, several independently wired regions.

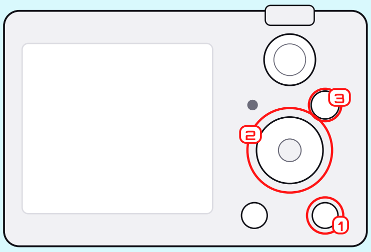
ADC ▶ MATRIX ▶ **SEGMENTS** ▶ PROFILES ▶ EXPOSURE ▶ JPEG + DNG

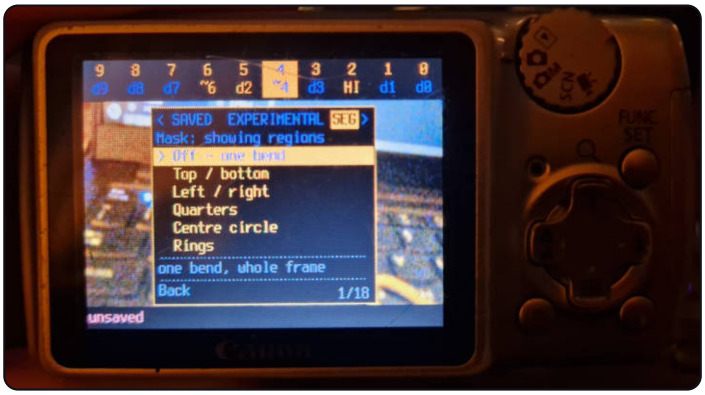
CUTTING THE FRAME INTO SHAPES

A segment layout gives each region of the frame its own matrix. Only the matrix is segmented - the profile chain models one frame buffer with one fault in it, so it still runs once, over everything, afterwards. That is why a checkerboard can have alternating colour logic and then share a single global tear.

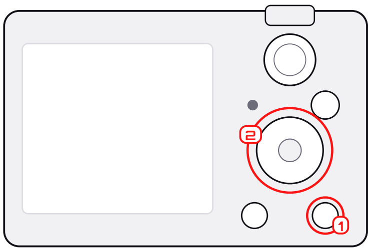
ON THE CAMERA

Patchbay ▶ MENU ▶ SEG. Pick a layout, pick region 1, choose a saved bend for it; the panel comes straight back to the region list for the next one. Back on the strip, an extra cell appears at the left end and selects which region the pins are editing.

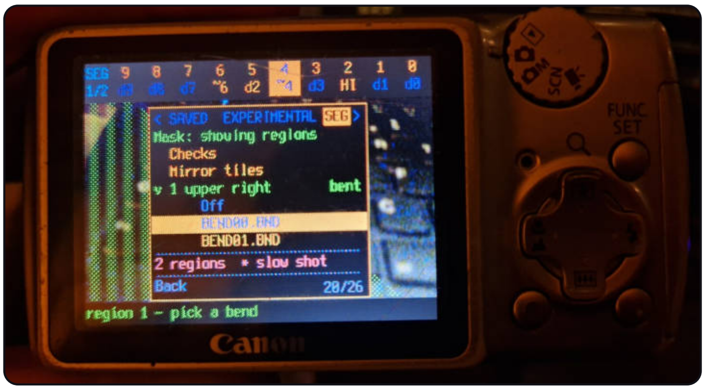




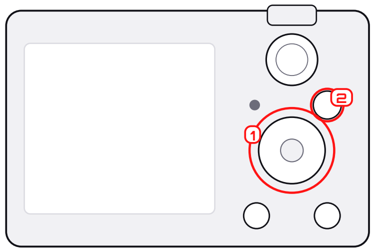
The SEG window



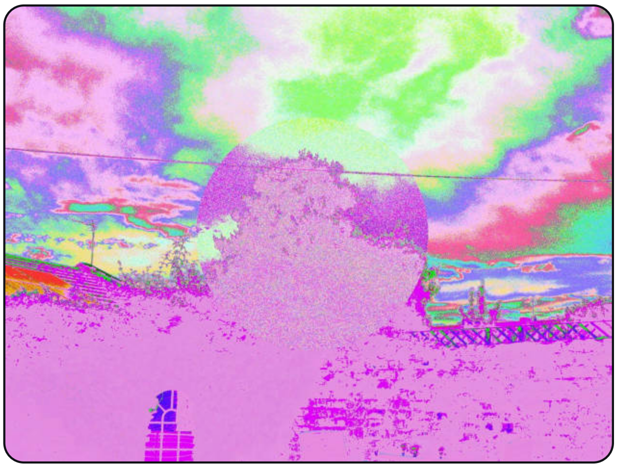
RIGHT again from EXPERIMENTAL. The layouts run from **Off** through the ten geometric ones to the six patterns. Off is a terminal row: it does not expand, because no segment layout is active, and the footer reads `segments off`.



Choosing a bend per region



Press a layout and its Size (where it has one) and regions open one level in, directly beneath it. Press a region and its saved presets open a second level in beneath that row. Here region 1 is `upper right` and marked `bent`. `2 regions * slow shot` warns that this one costs time at capture.

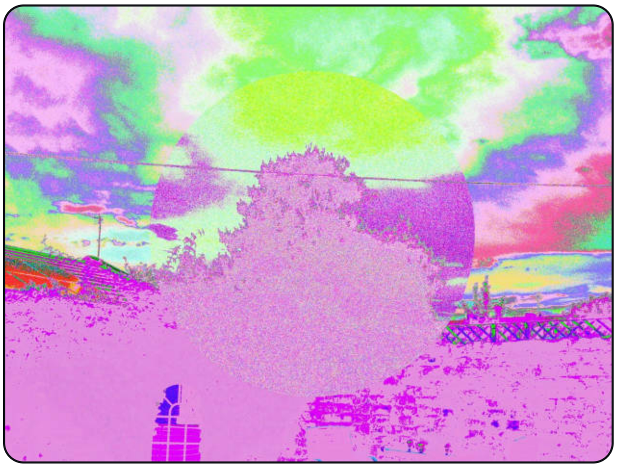


Matrix 10 data lines • random depth 3 • trash White

9	8	7	6	5	4	3	2	1	0
d9	d0	~7	d6	d5	~4	d3	d2	d1	d8

Segments **Centre circle** size 50%

Centre circle at 50%: the disc and everything outside it are wired differently, from two saved presets.



Matrix 10 data lines • random depth 3 • trash White

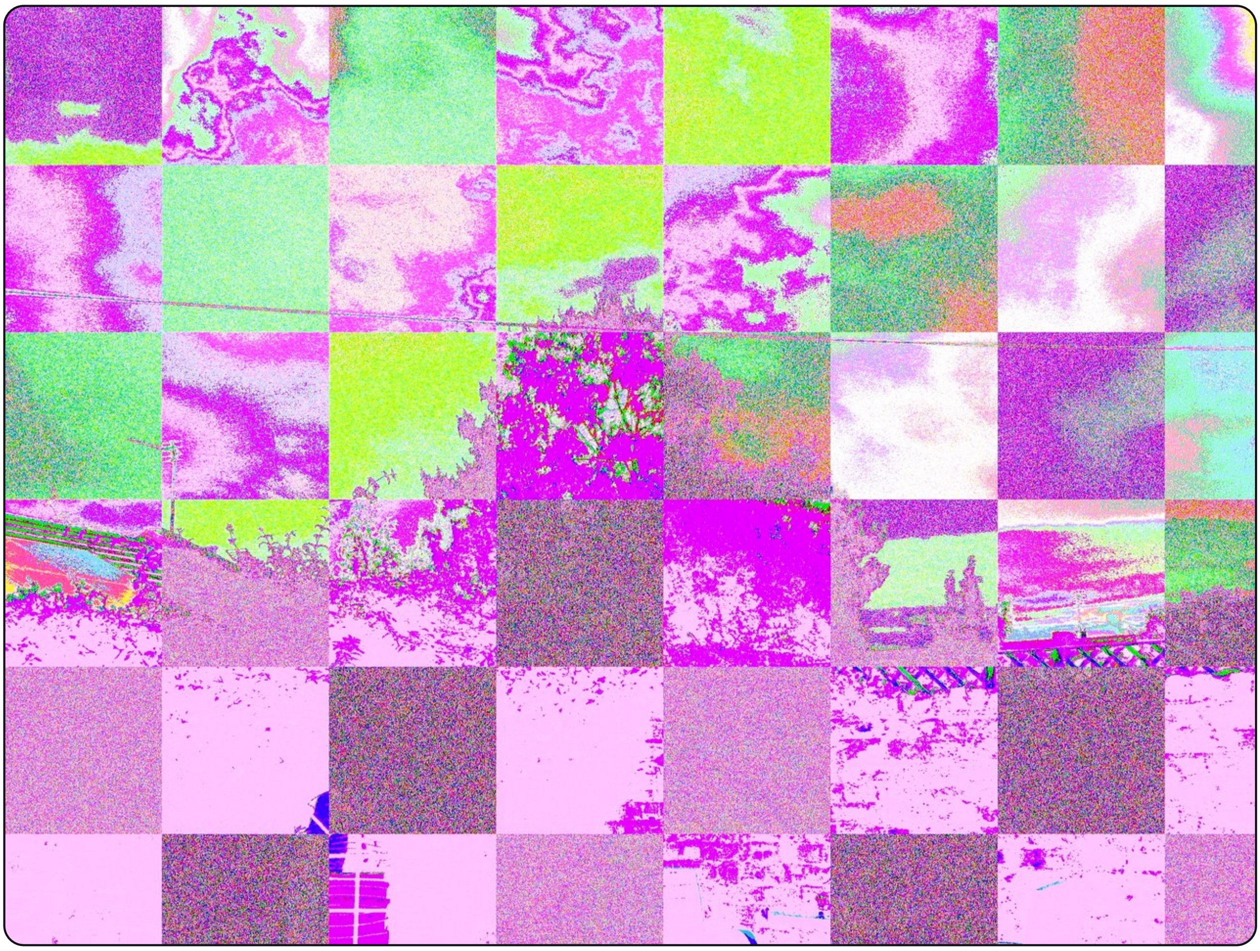
9	8	7	6	5	4	3	2	1	0
d9	d0	~7	d6	d5	~4	d3	d2	d1	d8

Segments **Centre circle** size 70%

The same two bends, the same scene, radius taken to 70%. Size is a knob, not a layout.

GOOD TO KNOW

A region is not a second photograph. Every region comes from the same shutter press and the same scene - only the routing of the ADC's numbers changes from place to place.



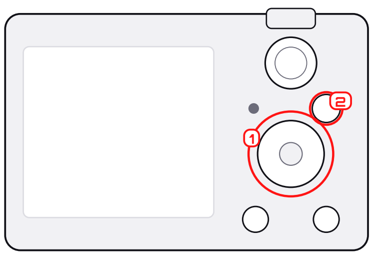
Matrix 10 data lines • random depth 3 • trash White

9	8	7	6	5	4	3	2	1	0
d9	d0	~7	d6	d5	~4	d3	d2	d1	d8

Segments **Checkerboard** size 70%

Checkerboard at 70%: four regions cycling across a grid, and the clearest demonstration in the manual that this is routing rather than a filter. The squares have no edges of their own and nothing is drawn over the picture - the numbers coming off the ADC simply take a different path depending on where on the sensor they came from.

Mask: showing regions



The most useful screen in the build. **Mask** draws the layout itself over the live view, so you can see which pixels belong to which region before spending a shutter press finding out. Here: `Diagonal`, region 1 upper right.

MORE DEMOS

These are from the **A460**.

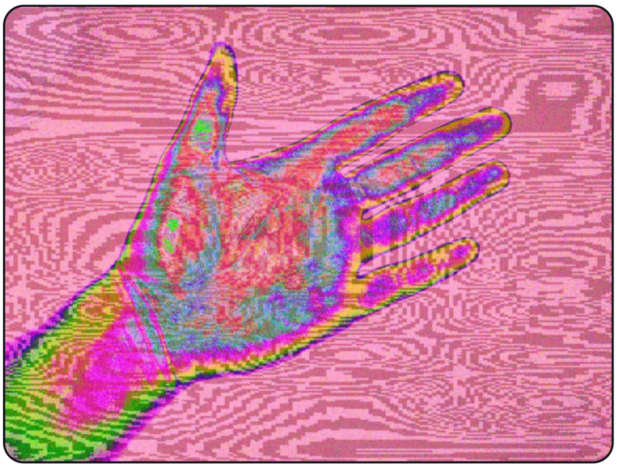
Matrix 10 data lines • random depth 3 • trash White

9	8	7	6	5	4	3	2	1	0
d9	d8	d5	d1	d7	d4	d3	d2	d0	d6

Segments **Warped lines** size 55%

Body Canon PowerShot A460

Warped lines across a face. The stripe field is noise-warped, so it follows nothing in the picture, and the collision between that and a face is the whole effect.



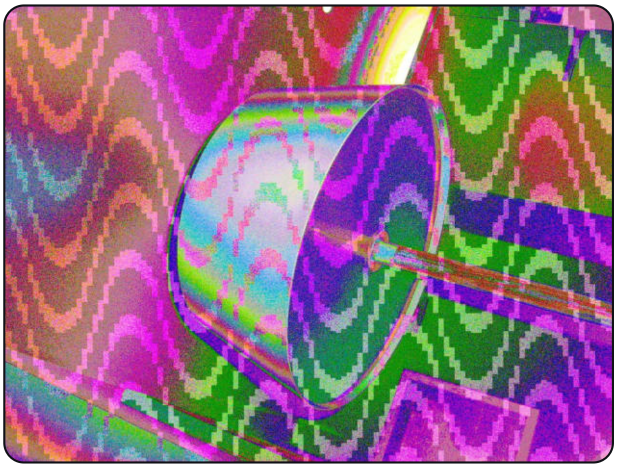
Matrix 10 data lines • random depth 3 • trash White

9	8	7	6	5	4	3	2	1	0
d9	d8	d5	d1	d7	d4	d3	d2	d0	d6

Segments **Warped lines** size 50%

Body Canon PowerShot A460

The same layout and the same two matrices on a hand, five points smaller. Two regions, and a boundary that is pure pattern.



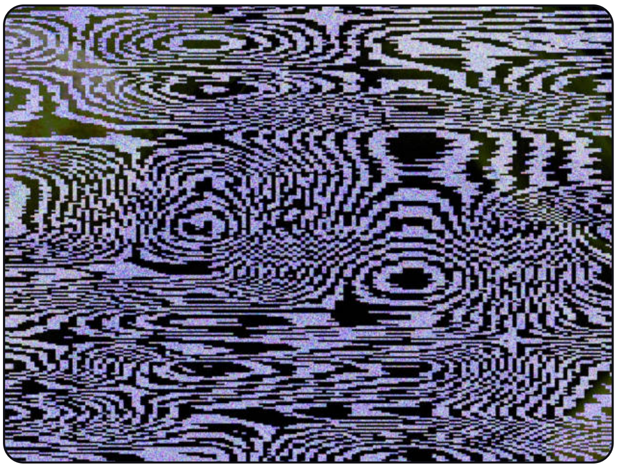
Matrix 10 data lines • random depth 3 • trash White

9	8	7	6	5	4	3	2	1	0
d9	d8	d5	d1	d7	d4	d3	d2	d0	d6

Segments **Wave lines** size 50%

Body Canon PowerShot A460

Wave lines on a lamp. The displacement is a sine, so the banding stays regular while the thing underneath it does not.



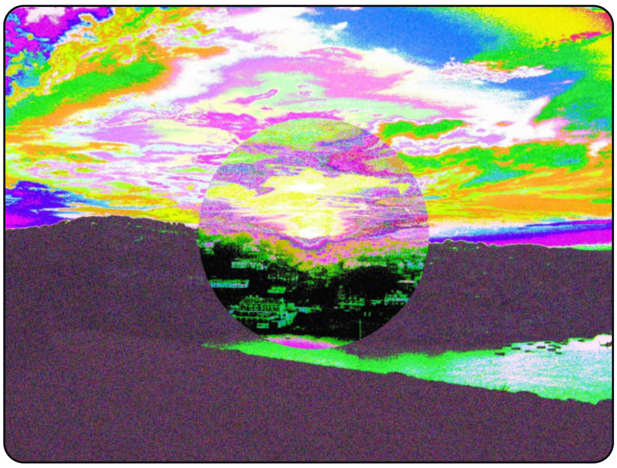
Matrix 10 data lines • random depth 3 • trash White

9	8	7	6	5	4	3	2	1	0
d8	d9	d7	d6	d4	d5	LO	d2	d1	d0

Segments **Warped lines** size 50%

Body Canon PowerShot A460

Warped lines with a figure and ground that are nearly opposites, which collapses the picture to two tones and leaves only the pattern.



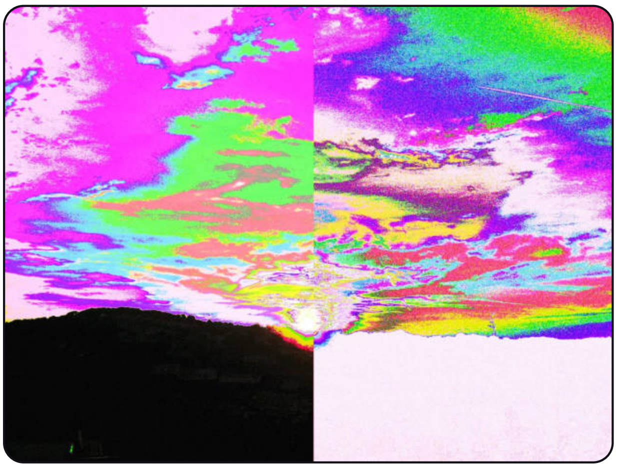
Matrix 10 data lines • random depth 3 • trash White

9	8	7	6	5	4	3	2	1	0
d9	d4	d6	d7	d5	d8	L0	d2	d1	d0

Segments **Centre circle** size 50%

Body Canon PowerShot A460

Centre circle: the disc gets one matrix and everything outside it another. A geometric layout holds still between frames, unlike the six pattern ones.



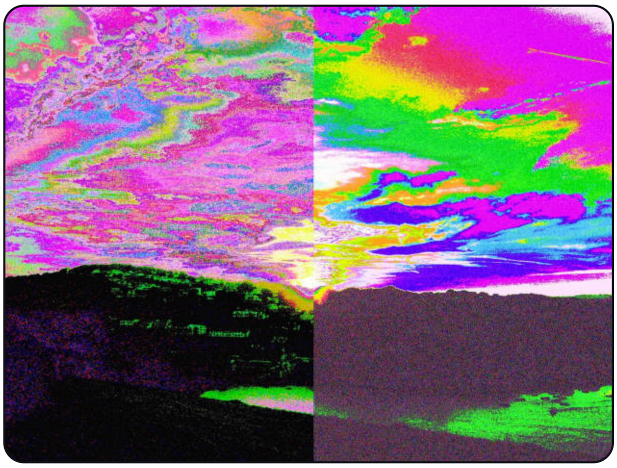
Matrix 10 data lines • random depth 3 • trash White

9	8	7	6	5	4	3	2	1	0
d6	d8	d7	d9	d5	d4	d3	~2	~1	d0

Segments **Left / right** size 50%

Body Canon PowerShot A460

Left / right, the simplest layout there is, and still enough to put two different skies in one photograph.



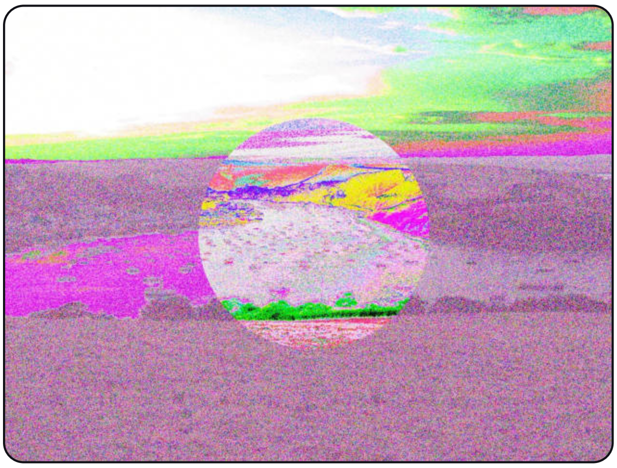
Matrix 10 data lines • random depth 3 • trash White

9	8	7	6	5	4	3	2	1	0
d9	d4	d6	d7	d5	d8	L0	d2	d1	d0

Segments **Left / right** size 50%

Body Canon PowerShot A460

The same left/right split with a darker ground. The seam is exactly down the middle of the sensor and nothing is drawn on it.



Matrix 10 data lines • random depth 3 • trash White

9	8	7	6	5	4	3	2	1	0
d4	d8	d7	d6	d3	d9	d5	d2	L0	d0

Segments **Centre circle** size 50%

Body Canon PowerShot A460

Centre circle again at the same radius, with the horizon running through the disc - the easiest way to see that both regions are the same photograph.

TRY THIS

Save two bends with clearly different palettes before you start, and use that pair for every two-region layout until you can read the shapes at a glance. Then replace one of them with something subtler.

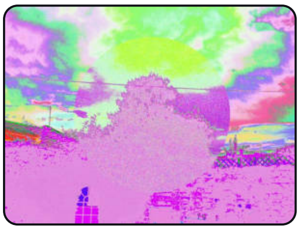
EVERY LAYOUT, ONCE

Sixteen layouts. The first ten are geometric and hold still; the last six are patterns, built from two bends - a figure and a ground - whose exact arrangement changes from frame to frame. Labels are read from each photograph's own recipe.

Top / bottom size 50%

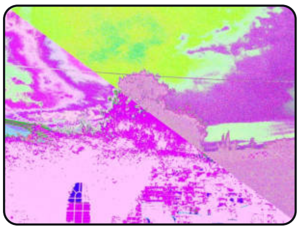
Left / right size 50%

Quarters size 50%

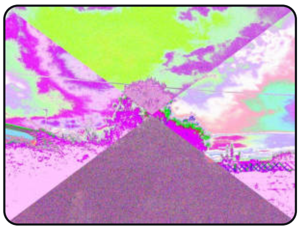


Centre circle size 70%

Rings size 70%

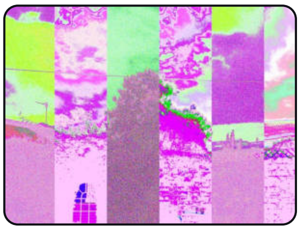


Diagonal size 70%

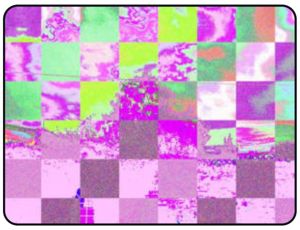


Wedges X size 70%

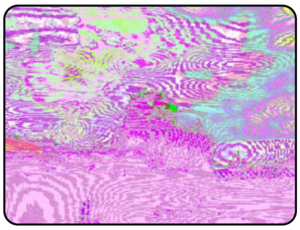
Bands size 70%



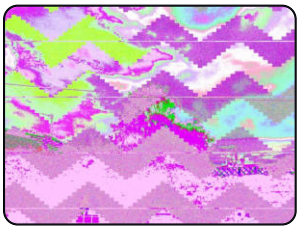
Bars size 70%



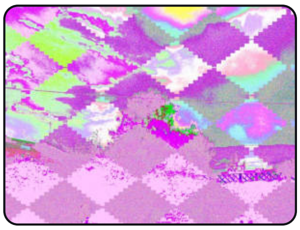
Checkerboard size 70%



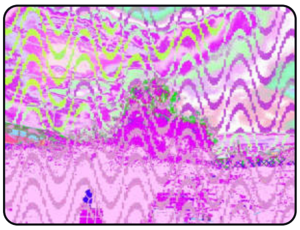
Warped lines size 70%



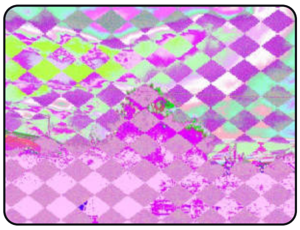
Zigzag size 25%



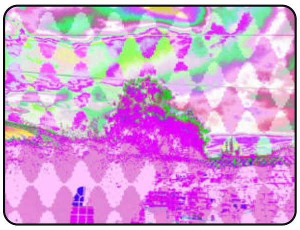
Truchet size 25%



Wave lines size 75%



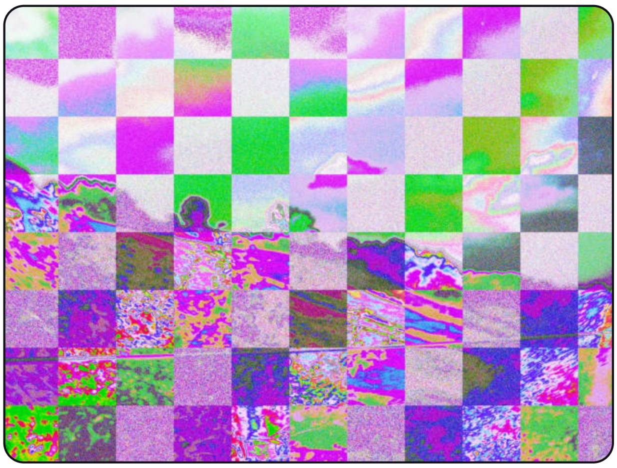
Checks size 75%



Mirror tiles size 75%

THE SAME LAYOUTS ON A TWELVE-BIT ADC

The A480 reads twelve bits per pixel rather than ten, so its patchbay is twelve pins wide and its strip has two more cells. Nothing else changes - the layouts, the sizes and the recipe format are the same, and a bend saved on one body is a legal bend on the other.



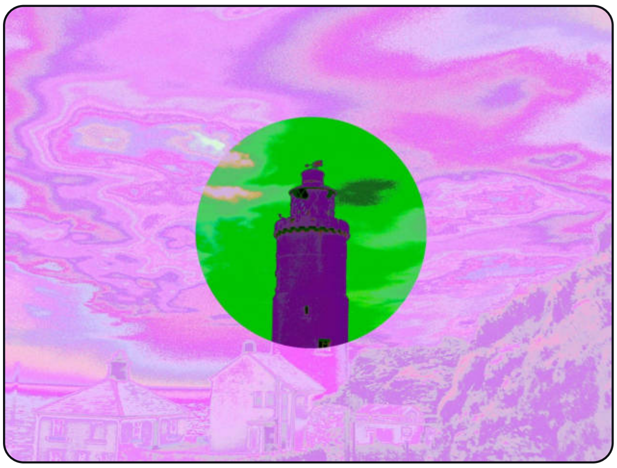
Matrix 12 data lines • random depth 5 • trash White

11	10	9	8	7	6	5	4	3	2	1	0
db	d8	d3	da	d7	d6	d5	d0	d9	d2	d1	d4

Segments **Checkerboard** size 50%

Body Canon PowerShot A480

Checkerboard at 50%, four regions cycling across the grid. Read the strip: twelve lines, and the two extra ones at the left are doing as much work as any of the others.



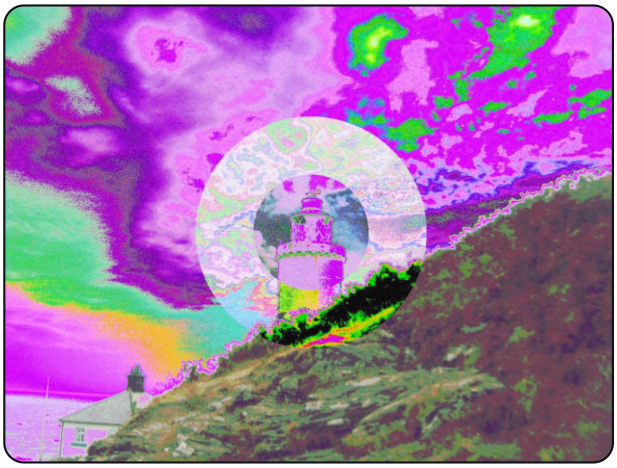
Matrix 12 data lines • random depth 5 • trash White

11	10	9	8	7	6	5	4	3	2	1	0
db	da	d0	d8	d9	HI	d5	d3	d4	d7	d1	d2

Segments **Centre circle** size 50%

Body Canon PowerShot A480

Centre circle on a lighthouse. One matrix inside the disc, another outside, and the subject deliberately placed across the boundary - the disc keeps the tower legible while the ground around it goes.



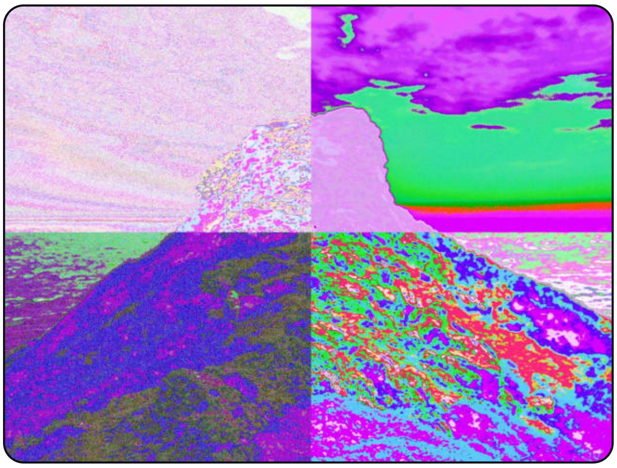
Matrix 12 data lines • random depth 5 • trash White

11	10	9	8	7	6	5	4	3	2	1	0
db	d8	d3	da	d7	d6	d5	d0	d9	d2	d1	d4

Segments **Rings** size 50%

Body Canon PowerShot A480

Rings - disc, annulus and outside, so three matrices in one frame from the same shutter press.



Matrix 12 data lines • random depth 5 • trash White

11	10	9	8	7	6	5	4	3	2	1	0
d6	d3	d1	~8	d7	db	d5	d4	da	d2	~9	d0

Segments **Quarters** size 50%

Body Canon PowerShot A480

Quarters with four unrelated recipes. Four regions is the maximum, and this is what the maximum looks like.

Size scales the pattern, and on the geometric layouts it is the radius or the band width. Press the layout to reveal its indented Size row, then use LEFT/RIGHT on the A470 or the rocker on bodies that have one.

CHAPTER FIVE

MULTIPLE EXPOSURE

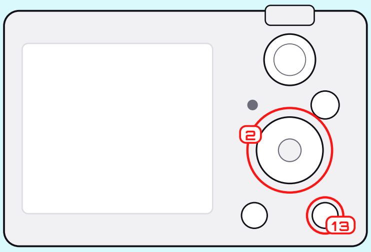
Several moments, developed into one finished image, in the camera.

ADC ▶ MATRIX ▶ SEGMENTS ▶ PROFILES ▶ **EXPOSURE** ▶ JPEG + DNG

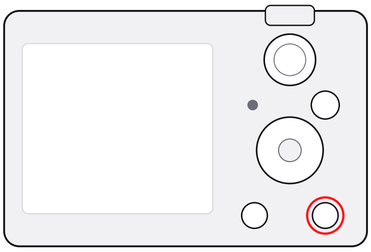
SETTING UP A SEQUENCE

ON THE CAMERA

CHDK menu ▶ **Multiple exposure.** Turn **Enable** on, choose a **Blend**, choose **2–9 exposures**, leave **Ghost on screen** on, and go back to shooting with MENU.

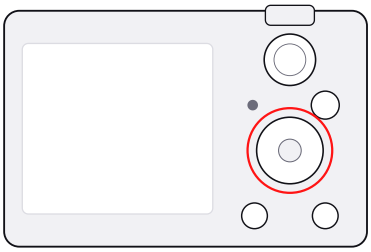


The CHDK menu



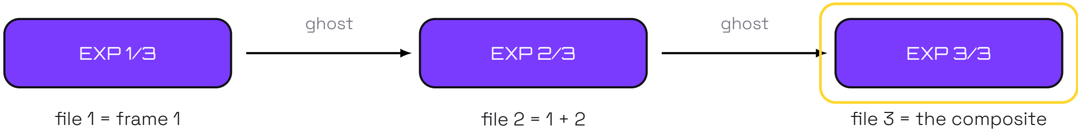
MENU outside the patchbay. **Circuit Bending** holds the settings the patchbay does not, and **Multiple exposure** is directly under it; **Game Boy emulator** is further down the same list.

Multiple exposure

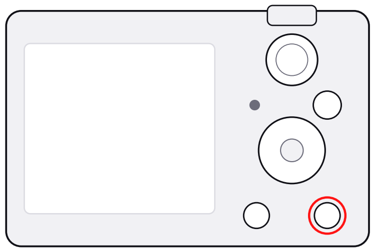


Enable, Blend, Exposures, Ghost on screen, Bend each exposure - and Cancel sequence in red at the bottom, the one thing that will not clear itself. LEFT/RIGHT change the value on the highlighted row.

The display reads `EXP 1/2` before the first frame. After each one, a five-level monochrome checkerboard ghost of what you have so far is held over the live view to compose the next against. The exposures live on the card between shots, in the ADC's own packed format: a raw frame is nine megabytes on this body and fifteen on the A480, and there is no room for either in memory.

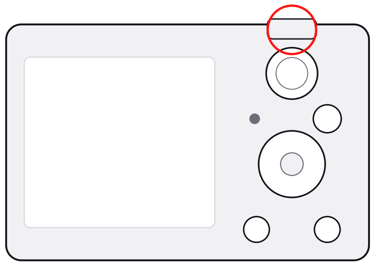


Waiting for the first frame



`1/2` at the top of the shooting screen: a two-shot sequence is armed and nothing has been taken yet. The patchbay strip is still up beside it, because the two are independent - you can repatch pins between exposures.

The ghost



After the first exposure, a monochrome checkerboard of what you have so far is held over the live view to compose the next against, and the counter reads `2/2`. This is what makes multiple exposure aimable rather than guesswork, and it is the one thing in this manual that prose cannot convey.

GOOD TO KNOW

Canon writes a JPEG after every shutter press, so the camera keeps every stage. In a three-shot sequence, file one is exposure one, file two is one plus two, and only the last file is the finished composite. This is not a bug and there is no way to suppress it - the picture you want is the last one.

ADDITIVE	Light sums; the black pedestal every exposure carries is subtracted back once, so a dark scene stays dark instead of drifting grey. Highlights clip. Best for light trails and bright objects against darkness.
LIGHTEN	The brighter pixel wins. Lights build over a dark scene without the background brightening at all.
DARKEN	The inverse: the darker pixel wins, so anything bright and transient disappears.
AVERAGE	Equal-weight ghosting with the exposure roughly unchanged. The classic double exposure, and the easiest place to start.

Bend each exposure decides the order. On, every ingredient is bent before it is combined. Off, clean frames are accumulated and both bend engines run once, on the finished composite. That one switch produces completely different pictures from the same scene. It is latched when the sequence starts, along with the blend and the frame count, so changing any of the three part way through affects the next sequence rather than this one.

SHOOTING THE LAYERS

Recipe none recorded

Exposure one. Frame the first subject and shoot; the camera then reads and writes a raw-sized temporary file, which takes a few seconds on hardware this old.

Recipe none recorded

Average composite. Move or rotate the camera between frames - the ghost is there to be argued with, not matched.

Recipe none recorded

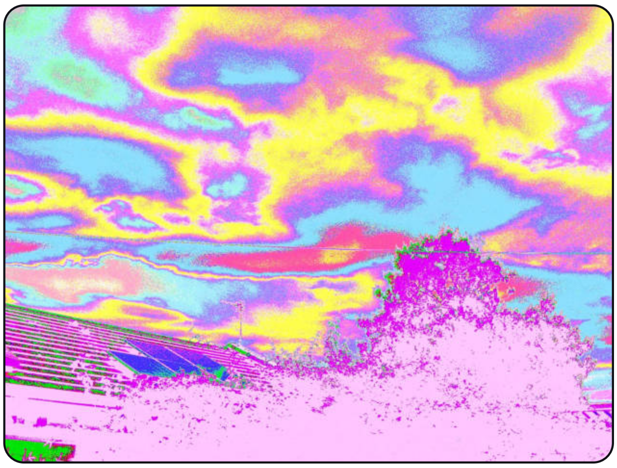
A pale second view over a dense first gives a soft, spectral overlay rather than a collision.

Recipe none recorded

A full rotation between exposures turns a familiar scene into a mirrored one.

A trailing exclamation mark on the counter - `EXP 1/2!` - means the sequence is still active but there is no ghost to paint; you can finish it regardless. Changing to playback, turning the mode dial or opening a menu does not cancel a sequence. Two things do: **Cancel sequence**, which also removes the temporary file, and switching **Enable** off. Power cycling forgets the sequence but leaves the file on the card until the next one overwrites it.

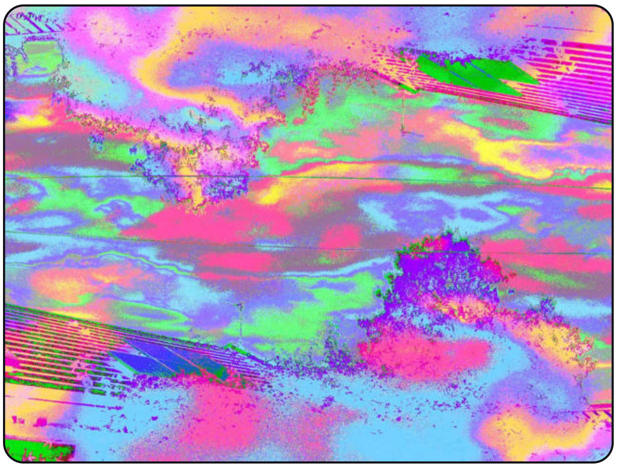
BENDS PLUS EXPOSURES



Matrix 10 data lines • random depth 3 • trash White

9	8	7	6	5	4	3	2	1	0
d5	d8	d7	**d9**	**d6**	d4	**d0**	d2	d1	**d3**

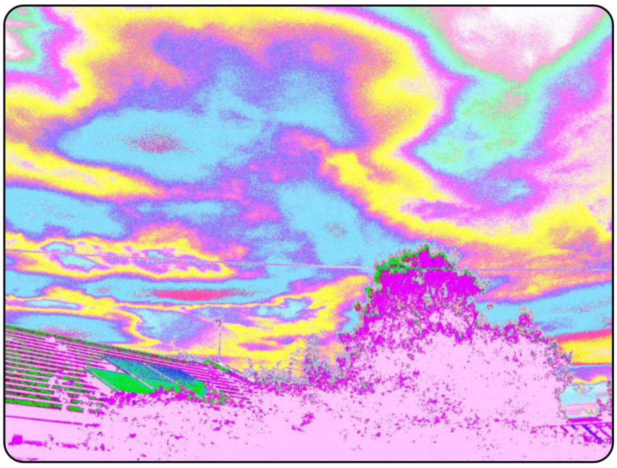
One bent ingredient. The matrix has already remapped sky and subject before anything is combined.



Matrix 10 data lines • random depth 3 • trash White

9	8	7	6	5	4	3	2	1	0
d5	d8	d7	**d9**	**d6**	d4	**d0**	d2	d1	**d3**

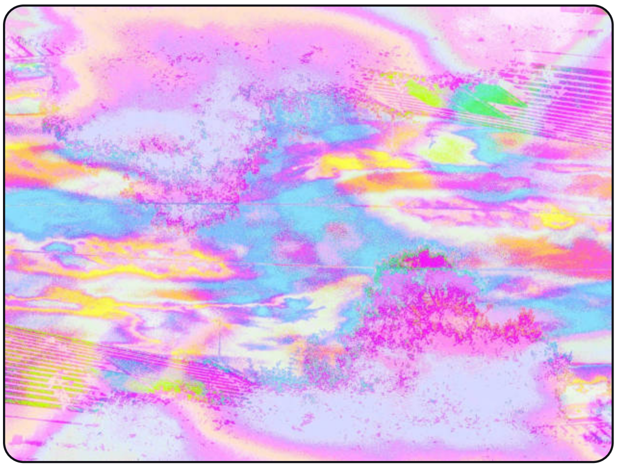
Layering bent frames compounds both colour and structure - the same recipe, arriving twice, out of register.



Matrix 10 data lines • random depth 3 • trash White

9	8	7	6	5	4	3	2	1	0
d5	d8	d7	**d9**	**d6**	d4	**d0**	d2	d1	**d3**

A shifted second exposure introduces a second contour of the same remapped palette.

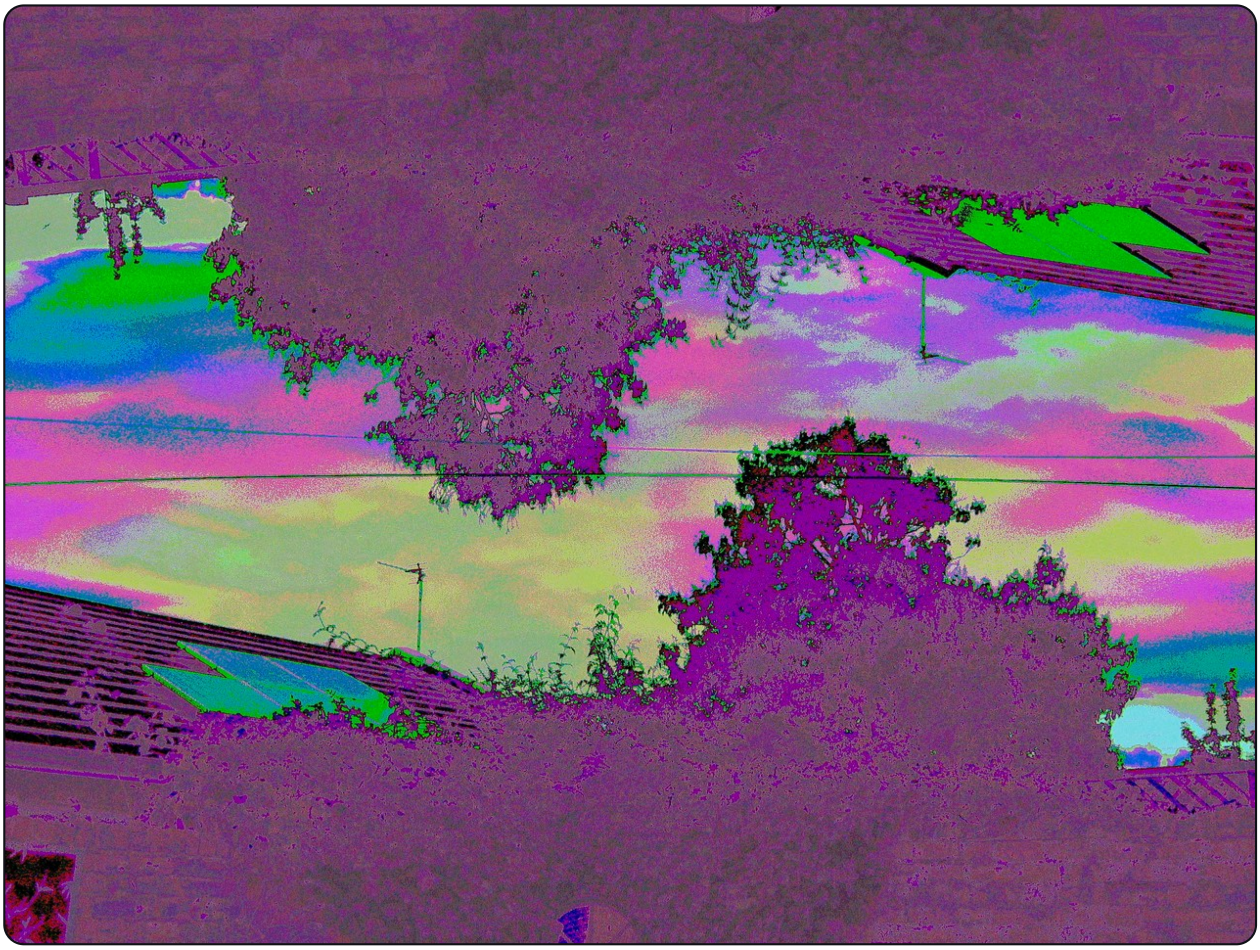


Matrix 10 data lines • random depth 3 • trash White

9	8	7	6	5	4	3	2	1	0
d5	d8	d7	**d9**	**d6**	d4	**d0**	d2	d1	**d3**

The finished composite: a dense field where the original scene survives only as structure.

WHAT IT CAN LOOK LIKE



Recipe none recorded

Several exposures of the same wall, bent, layered and left to collide. This is the far end of everything in this manual - two engines and a multiple exposure on one shutter press - and it is also a cautionary tale, because it carries no recipe. It was shot with **Record bend with shot** switched off, so the routing that made it is gone and this photograph cannot be made twice. Everything else printed in this manual can be, which is the entire argument for leaving that setting on.

For control, save a bend and keep it loaded through the whole sequence, as these four were. For unpredictability, leave **Bend each exposure** on and reroll the matrix between shutter presses. The recipe stored in the final JPEG is the bend state as it stood when that frame was processed, so if you deliberately change recipes mid-sequence, the file records the last one only - keep notes.

CHAPTER SIX

GAME BOY

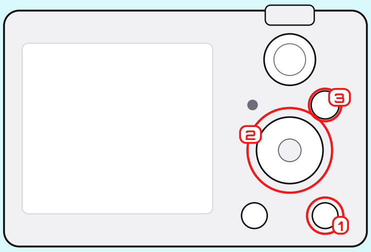
Yes, the camera is also a handheld console. And the 1998 camera cartridge looks through the Canon lens.

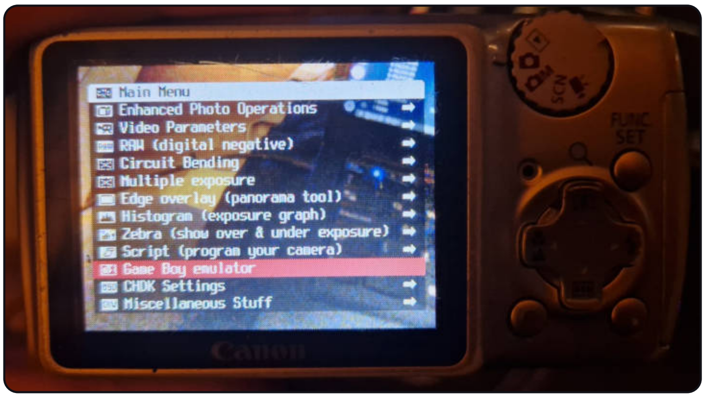
ADC ▶ MATRIX ▶ SEGMENTS ▶ PROFILES ▶ EXPOSURE ▶ JPEG + DNG

STARTING A GAME

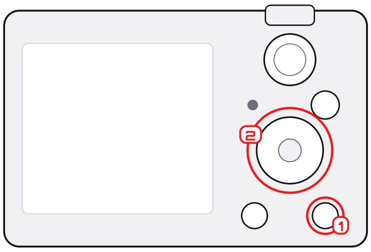
ON THE CAMERA

CHDK menu ▶ Game Boy emulator. UP/DOWN pick a ROM, SET runs it, MENU quits safely.



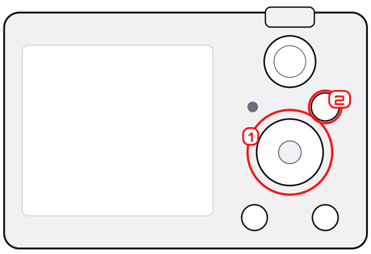


Finding it



Game Boy emulator in the CHDK menu. It is a module, so it is loaded off the card when you open it rather than sitting in memory the whole time.

The ROM picker



Whatever is in `CHDK/GBC`, listed. SET runs the one under the cursor, MENU leaves. ROMs are not supplied with the build - these two are the owner's.

Put legally obtained `.gb` or `.gbc` files in `CHDK/GBC`, unzipped - archives are not scanned. Battery-backed games write ordinary `.SAV` files into `CHDK/GBC/SAVE` when the game commits, on a dirty timer, and when you quit.

ARROWS	Game Boy D-pad
SET	A
HALF-PRESS SHUTTER	START
TAP PRINT	SELECT - sent when the key is released, because a held PRINT means something else
HOLD PRINT	Switch between PAD and LENS mode
MENU	Quit to the ROM picker. Deliberately not remappable: a game that swallowed it would strand you.
DISP	B. See below: the A470 has no DISP key, so B needs setting up there first.

GOOD TO KNOW

The A470 has no B button out of the box. B is the DISP key, and the A470's ten keys do not include one. The fix is in CHDK: **Miscellaneous** ▸ **Extra button** ▸ **Display** remaps the **power button** to act as DISP, which is then B. Without it, games that need B are unplayable - though the Game Boy Camera itself only needs A. The A410, A460 and A480 all have that key already and need none of this.

GOOD TO KNOW

There is no Game Boy sound. These PowerShots expose whole Canon sound clips, not a stream of audio samples a console core could feed.

LENS MODE

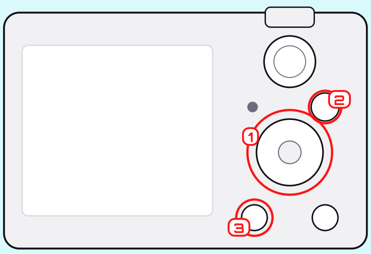
Holding PRINT for 600ms swaps what the arrows mean. In `LENS` mode, UP/DOWN drive the real Canon zoom and LEFT/RIGHT change the screen tint; SET and START keep working, so the Game Boy Camera can still shoot after you have zoomed. The strip down the left of the screen says which mode you are in. Hold PRINT again to get the D-pad back.

A CAMERA INSIDE A CAMERA

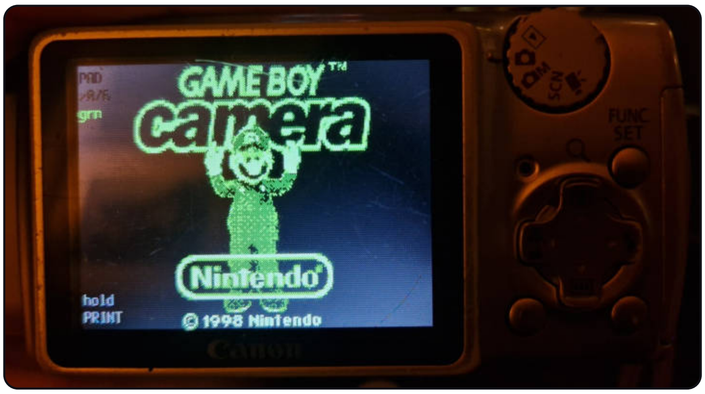
The Game Boy Camera ROM is not handed a screenshot. Its emulated cartridge - a MAC-GBD mapper and an M64282FP analogue image sensor - pulls live luminance from the Canon viewport and then runs the real cartridge model over it: exposure, edge enhancement, four-tone dithering, tile packing. Taking a picture inside the ROM photographs whatever the Canon lens is pointed at, in 128×112 Game Boy pixels.

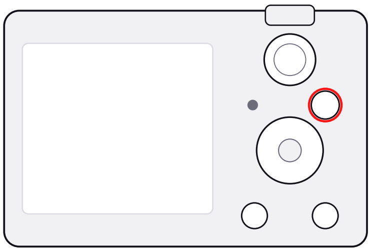
ON THE CAMERA

Put your legally obtained Game Boy Camera ROM in `CHDK/GBC` and run it. Enter **Shoot** inside the ROM, frame with the Canon lens, and press SET - the Game Boy's A button. Hold PRINT for `LENS` mode if you need the zoom.



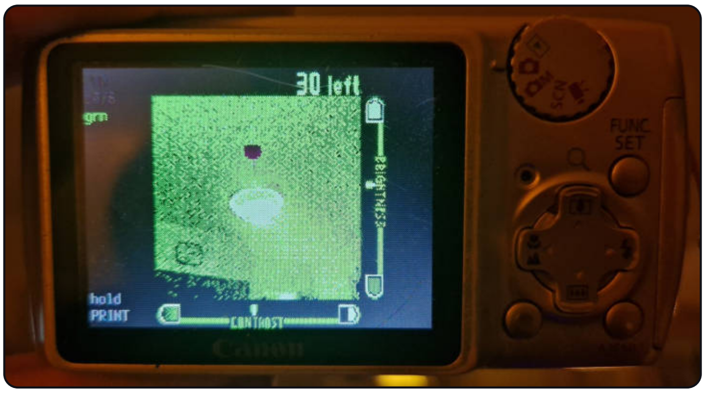
The cartridge, running

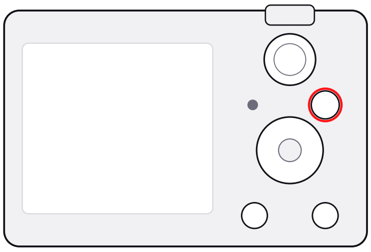




A 1998 ROM booting on a 2008 PowerShot. The strip down the left belongs to this port, not the cartridge: it says which mode the arrows are in (`PAD`), where the Canon zoom is (`z3.6`) and what tint the screen is using (`grn`).

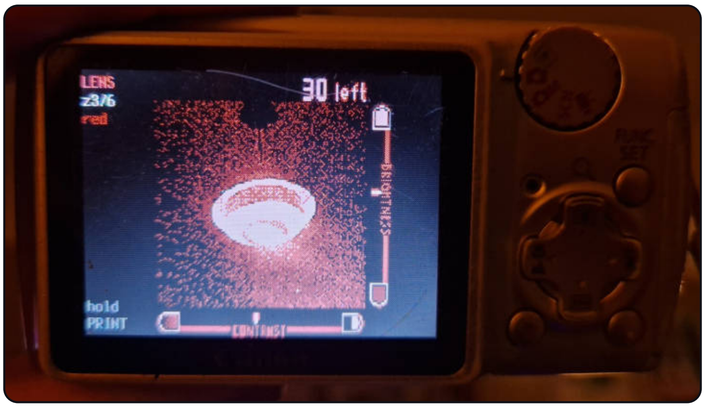
Shooting, through the Canon lens

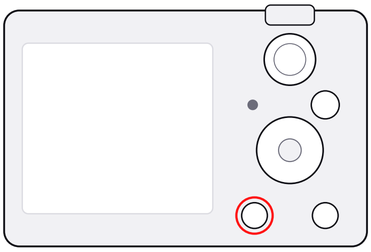




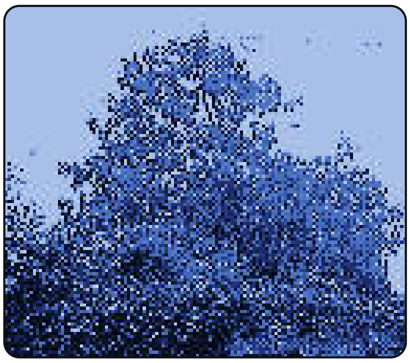
Inside the ROM's Shoot mode. What the cartridge is metering is live luminance from the Canon viewport, put through the real M64282FP sensor model - exposure, edge enhancement and four-tone dithering included. `30 left` is the album's remaining slots.

LENS mode



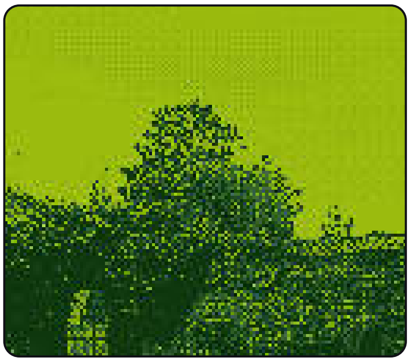


Hold PRINT for 600 ms and the arrows change job: UP/DOWN drive the real zoom, LEFT/RIGHT the screen tint - here from `grn` to `red`. The strip says `LENS` so you can tell which mode has the d-pad. Hold PRINT again to give it back.



Pictures are written out as viewable bitmaps to `DCIM/GBC/GBC_nnnn.BMP`, numbered and never reused, so they come off the card with the rest of the shoot. They also stay in the cartridge's own save data, and a slot-named copy goes to `CHDK/GBC/PHOTOS`. The cartridge album holds thirty and the ROM reuses those slots, which is what the numbered DCIM copies exist to survive. The album is also kept short on purpose: once it holds more than four, the oldest entries are retired, so the ROM's own counter sits near its maximum instead of walking down to zero and refusing to shoot. It only ever retires a picture this session has already written to `DCIM`, so nothing is freed that is not already on the card. Quit with MENU and wait for the writes to finish before pulling the card.





TRY THIS

A good first session. Shoot a high-contrast subject in even daylight. If it is too small, hold PRINT for LENS mode and zoom with UP. Change tint with LEFT/RIGHT - that is the display palette, not the four-tone data being saved. Return to PAD mode before navigating the ROM's menus. Keep the picture inside the cartridge, then quit with MENU so it is exported.

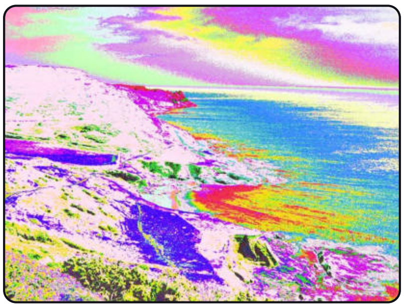
These are shown enlarged with hard edges on purpose. The square pixels are the picture, not a low-quality preview of one.

THE OTHER THREE BODIES

The build runs on seven PowerShots - the A410, A430, A460, A470, A480, A540 and A640 - and this manual's photographs come from four of them. They do not behave identically. All of them are ten-bit except the A480, which is twelve. Sensor size, lens and noise differ, and so does what a given recipe looks like coming out of them. The JPEG bus is the one profile that is not live on all of them: it needs an address recovered per firmware, and so far that is the A470 and the A480. Everything below is one of the other three bodies, labelled with the body and the recipe it recorded.

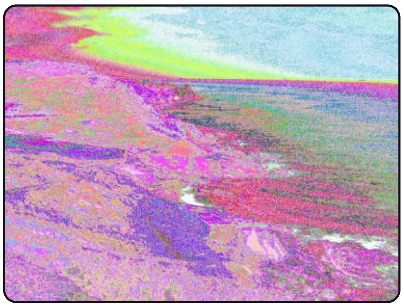
9 8 7 6 5 4 3 2 1 0
d9 d8 d7 LO HI d4 d3 d2 d0 LO

depth 6 · whole frame
Canon PowerShot A410



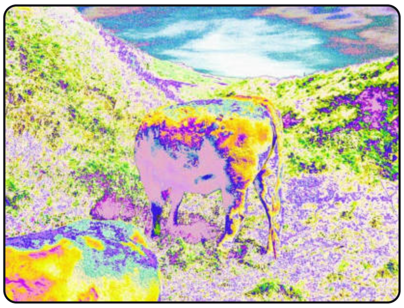
9 8 7 6 5 4 3 2 1 0
d6 d8 d9 d2 d5 ~4 d0 d1 d7 d3

depth 6 · whole frame
Canon PowerShot A410



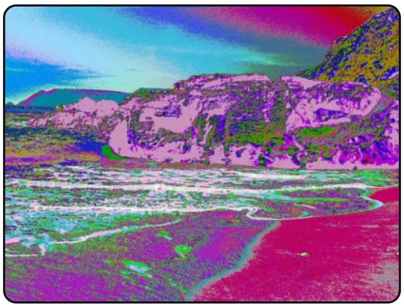
9 8 7 6 5 4 3 2 1 0
d9 d4 ~7 d2 d5 d8 d3 HI d1 d0

depth 6 · whole frame
Canon PowerShot A410



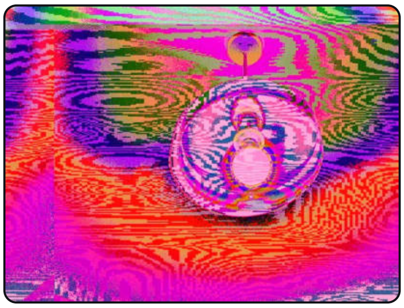
9 8 7 6 5 4 3 2 1 0
d7 d5 d9 d2 d1 ~3 d4 d6 d8 d0

depth 6 · whole frame
Canon PowerShot A410



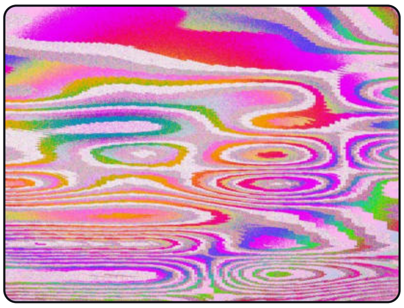
9 8 7 6 5 4 3 2 1 0
d9 d8 d5 ~6 ~7 d2 d1 d4 d0 d3

depth 6 · whole frame
Canon PowerShot A410



9 8 7 6 5 4 3 2 1 0
d9 d5 d7 d1 d8 d3 d4 d2 ~0 d6

depth 6 · Warped lines size 50%
Canon PowerShot A410

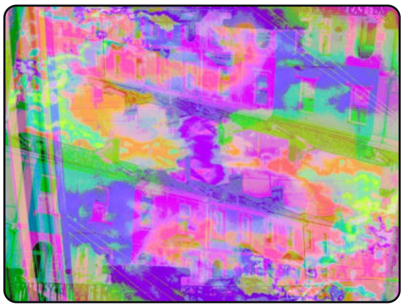


9 8 7 6 5 4 3 2 1 0
d9 d6 ~7 d8 d5 d4 d3 d2 HI d0

depth 3 · Warped lines size 50%
Canon PowerShot A460

9 8 7 6 5 4 3 2 1 0
d6 d8 d7 d9 d5 d4 d3 ~2 ~1 d0

depth 3 · Left / right size 50%
Canon PowerShot A460

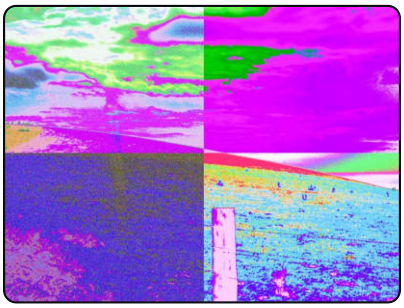


9 8 7 6 5 4 3 2 1 0
d9 d6 d7 d8 d5 d4 d2 d3 ~1 d0

depth 3 · whole frame
Canon PowerShot A460

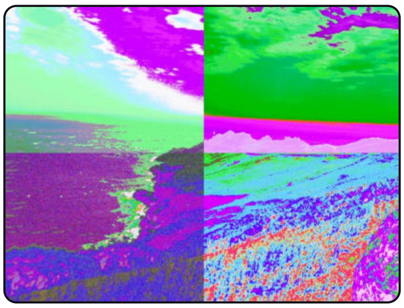
9 8 7 6 5 4 3 2 1 0
d9 d6 d7 d8 d5 d4 d2 d3 ~1 d0

depth 3 · **whole frame**
Canon PowerShot A460



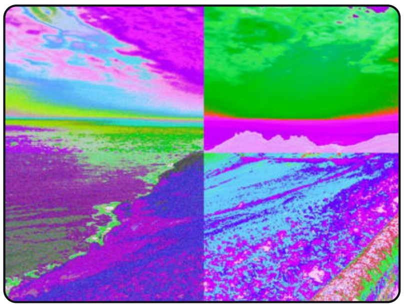
11 10 9 8 7 6 5 4 3 2 1 0
db d8 d3 da d7 d6 d5 d0 d9 d2 d1 d4

depth 5 · **Quarters** size 50%
Canon PowerShot A480



11 10 9 8 7 6 5 4 3 2 1 0
da LO d9 d8 d1 d0 d5 d4 LO d2 d7 d6

depth 5 · **Quarters** size 50%
Canon PowerShot A480



11 10 9 8 7 6 5 4 3 2 1 0
d0 da ~7 d9 d8 LO d5 d4 d3 d2 d1 db

depth 5 · **Quarters** size 50%
Canon PowerShot A480

11 10 9 8 7 6 5 4 3 2 1 0
db da d0 d8 d9 HI d5 d3 d4 d7 d1 d2

depth 5 · **Centre circle** size 50%
Canon PowerShot A480

Two things are worth reading off this page. The A410 frames are all at random depth 6, the A460 ones at 3 and the A480 ones at 5 - depth is how many of the ten or twelve leads the randomiser moves, and it is the single control that most changes how far from a photograph you land. And every segmented frame here is doing what the segments chapter described, on a different sensor, from a recipe file any of these bodies can read.

MAKING IT YOURS

None of this changes a photograph. It changes what the camera looks and sounds like while you are taking one, and all of it is on the card rather than in the build.

THEMES

NEW **CHDK settings ▸ Visual settings ▸ Theme** recolours everything this build draws - the patchbay, the record-screen menus and the overlay - in one move.

APPLE	The six stripes of the old logo and nothing else. No white and no grey anywhere.
VAPORWAVE	Pink, turquoise, purple and yellow, with no green.
FRUTIGER AERO	Turquoise glass, deep blue, grass green, silver.
TERMINAL	One hue at three brightnesses.
GUNMETAL	Solid grey, silver and white. Ordinary and inactive text use the camera's opaque grey rather than its translucent grey entries, so unselected bends, titles, separators and the `JPG`, `X off` and `SEG off` readouts stay visible against the smoked backing.

A theme is an assignment, not a palette: the palette belongs to Canon, so a theme chooses which of the colours the camera already has carries each meaning. The per-element colour pickers under the Theme row still work and still win, so a theme sets defaults rather than locking anything.

Two things deliberately ignore the theme. The patchbay's pin colours stay as they are, because ten traces crossing in a hundred and fifty pixels are only followable while no two look alike. And the splash credit stays white, which is the lightest colour these bodies have.

GRIDS AND RETICLES

NEW **Grid settings ▸ Load grid from file** reads `CHDK/GRIDS`. Alongside CHDK's own, this build adds `bend_cross`, `bend_thirds` and `bend_safe` for framing, and a set of gun-sight reticles - red dot, green dot, holographic ring, a 4x scope with a bullet-drop ladder, a thermal bracket and a mil-dot crosshair. They are written in CHDK's portable coordinate space and name their colours, so they draw correctly on every body here.

SOUNDS

NEW Put `shutter.wav`, `button.wav` or `selftimer.wav` in `CHDK/SOUNDS` and the camera uses them instead of Canon's. Supply one, two or none - a file that is missing or that the camera will not accept leaves the stock clip in place, so there is no way to end up with a silent shutter by mistake.

The format is narrow because Canon's own sound slots are: **RIFF/WAVE, 11025 Hz, unsigned 8-bit, mono**. Anything else is rejected rather than played badly. The startup sound is not on this list - it plays before the card is mounted, so on the bodies that have a custom one it is built in rather than read.

GOOD TO KNOW

Sound support is per body. It is wired and working on the A470, A460 and A480; on the A410 there is no My Camera sound table to point at, and on the A430 the startup route reloads Canon's own clip. A card with sounds on it does no harm on any of them.

THE BOOT SCREEN

NEW On the **A480**, **CHDK settings ▸ Boot screen ▸ Import JPG/PNG** opens a file browser at the root of the card. Pick any `.jpg` or `.png` under 2 MB and the camera fits it inside 320x240 centred, with black bars where the shape does not match, re-encodes it into the startup format Canon's own decoder expects, and writes the result into `DISKBOOT.BIN` itself. The build it replaces is kept beside it as `DISKBOOT.BAK`. Power cycle to see it.

Quality is not a setting: the encoder starts high and steps down until the frame fits the 18426 bytes Canon reserves for its logo. A picture too detailed to fit even at the lowest step is refused rather than installed badly, and so is anything the decoder will not read. Flat artwork with large areas of one colour survives that squeeze far better than a photograph does - which, on a camera whose whole subject is what happens to photographs, is worth knowing before you pick one.

On every other body the screen is built rather than imported. `tools/mkbootimg.py` turns a 320x240 JPEG into the header the port compiles in, so changing it there means rebuilding rather than copying a file to the card.

GOOD TO KNOW

This works by replacing Canon's own startup image, in the buffer the My Camera code copies it into, in the moment between that copy and the firmware displaying it. A body that shows no startup screen of its own has no such buffer to overwrite, so there is nothing to replace and the customisation simply does nothing. That is not a failure and nothing else about the build changes - the camera boots as it always did, straight to the overlay.

CHAPTER SEVEN

FIELD NOTES

How to get deliberate accidents instead of only accidents.

ADC ▶ MATRIX ▶ SEGMENTS ▶ PROFILES ▶ EXPOSURE ▶ JPEG + DNG

A WORKFLOW THAT REPEATS

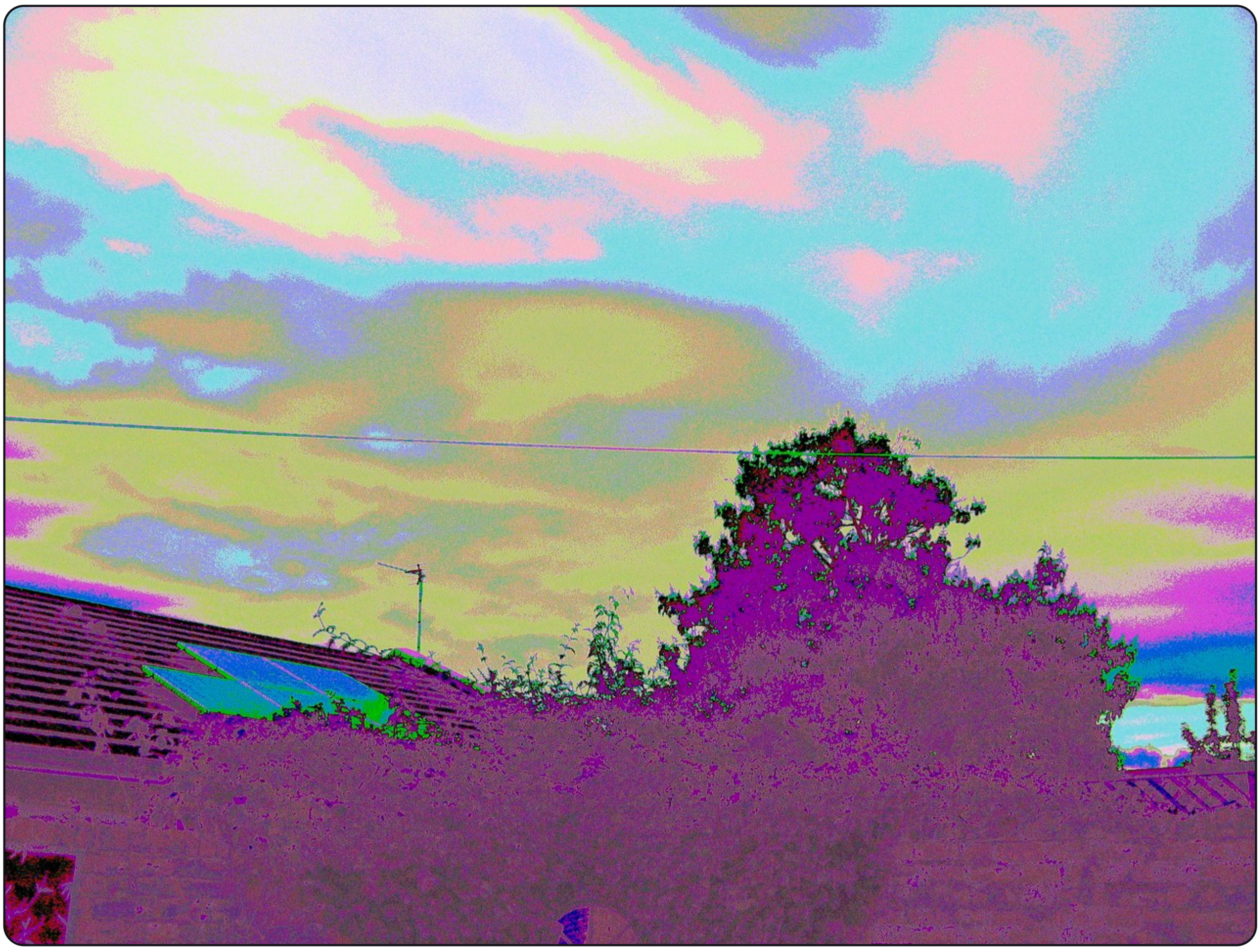
1. Take a straight control frame.
2. Change one system - matrix, profile, segments, or exposures.
3. Shoot it at a low setting and a high setting without moving.
4. Review. When one works, load its recipe from the image straight away.
5. Save it to a numbered slot, then mutate one edge for variations.
6. Copy the whole `DCIM` folder after the session, with the JPEG comments intact.

SUBJECTS THAT BEND WELL

Hard horizons. Bare tree branches. Signs and lettering. Faces against an empty wall. Repeating windows. Water. Clouds with structure. Point lights at night. Anything with a silhouette you would recognise in one frame of a film.

WHEN IT TURNS TO MUSH

Reset to straight and restore the top one or two pins first; lower the random depth; remove the last profile from the chain; or reduce the number of different recipes in the segment layout. A glitch reads as stronger, not weaker, when something recognisable survives it.



Matrix 10 data lines • random depth 3 • trash White

9	8	7	6	5	4	3	2	1	0
d9	d8	d7	~6	d2	d4	d3	HI	d1	d0

Pin 6 inverted, pin 2 tied `HI`, pin 5 fed from line 2 - three edges, and the top of the bus, pins 9 to 7, left completely alone. The building is still a building. That restraint is the whole technique.

TROUBLESHOOTING

FROZEN AFTER A SHOT	Look for the blue activity light. Bus sources, starred profiles, long chains and multiple exposure all take real time on this hardware. Wait. Do not power off.
NO VISIBLE BEND	Check the matrix is not straight, that the region you edited is the active one, and that the profile is actually enabled rather than only selected.
THE SAVED NAME VANISHED	You edited the recipe, so it is correctly marked `unsaved bend`. Save it to a new slot if you want to keep it.
CANNOT LOAD FROM A JPEG	The comment was probably stripped by an editor or an export. Try the original file off the card, or its `.BND` sidecar.
THE NEXT SHOT JOINED AN OLD ONE	A multiple-exposure sequence is still running. **Cancel sequence**.
NO GAME BOY ROMS FOUND	Copy unzipped `.gb` / `.gbc` files into `CHDK/GBC`. Archives are not scanned.
NO GAME BOY B BUTTON	Miscellaneous ▸ Extra button ▸ Display, which puts B on the power button.
GAME BOY CAMERA PICTURE NOT EXPORTED	Quit with MENU rather than pulling the card, wait for the writes, and check free space.
I WANT A NORMAL CAMERA BACK	Power off and use an ordinary, non-bootable SD card. Nothing was written to the body.

If a real fault survives a power cycle with a normal card, take the batteries out for a minute and test the stock camera. Do not format the card until the photographs and `CHDK/BENDS` are backed up.

CREDITS AND SOURCES

CHDK - community project, GPL source: `https://chdk.fandom.com/wiki/CHDK` and `https://github.com/CHDKUtil/CHDK`.

Game Boy emulator - `koenk/gbc`, MIT licence, pinned in the source tree. The Game Boy Camera mapper and sensor model derive from AntonioND's `gbcam-rev-engineer`, GPL-3.

Photographs - Rewired Optics, shot on the A470 this manual describes. Page-edge backgrounds are from the public gallery at `https://www.rewired-optics.com`. Every plate's state strip and every catalogue label was decoded from that frame's own `BSH1` recipe by `tools/gen_manual_assets.py`; nothing was assigned by eye.

Type - Display face **Mignova** by Sipanji, bundled with the project under a freeware, non-commercial licence: obtain the creator's commercial licence before selling or commercially distributing this manual. Headings **Orbitron** by Matt McInerney and **Michroma** by Vernon Adams; text **Space Grotesk** by Florian Karsten; machine text **Space Mono** by Colophon Foundry. All four are SIL Open Font Licence 1.1; the licences are in `assets/fonts/`.

GO BREAK SOME LIGHT

@rewired_optics